

SUBJECT: Web Programming**YEAR:** First Year**SEMESTER:** I**ACADEMIC YEAR:** 2026-27

Practical Index

Practical No.	Module No.	Title of the Experiments	Type of Experiment		Topics to be highlighted	CO Map
			PBL	Newly Added		
1	1	Use of Basic and Advanced text formatting Tags, Lists			Text Formatting, Lists	
2	1	Creating Hyperlinks, Tables, semantic tags			Hyperlinks, Tables, Semantic Tags	
3	1	Formatting Text and Paragraph by Using Style Sheets and displaying graphics			Inline, Internal, External Style Sheets, Layout & Graphics	
4	2	HTML Form designing using different controls			Form Design and Input Controls	
5	2	Basic Java script programs, Java script for validating user fields.			Variables, Functions, Events, Form Validation	
6	2	PHP scripts to perform simple mathematical operations and Retrieving data from HTML forms			Variables, Operators, Form Handling	
7	3	PHP to work with Arrays and Files (Reading / Writing).			Indexed Arrays, Associative Arrays, File Reading/Writing	
8	3	Advanced PHP: sessions and cookies, filters.			Sessions, Cookies, Filters	

9	3	PHP and MySQL: Create a database and tables.			Database Creation, Tables, SQL Commands	
10	4	PHP and MySQL Integration: Perform CRUD Operations (Insert, Update, Delete, Display)			CRUD Operations	
12	1,2,3,4	Mini Project Development: Develop a simple dynamic web application			Dynamic Web Application Development	

PRACTICAL 1

1. Aim

Create a web page for "Emerging Technologies Information Portal" using HTML

2. Prior Concepts

- Basic HTML structure
- Use of web browser and text editor

3. New Concepts

- Formatting tags: , <i>, <u>, , , <p>, <h1> to <h6>
- Advanced tags: <mark>, <sup>, <sub>, <s>, <ins>,
- Lists and their attributes: , , <dl>

4. Procedure

- Open a text editor (VSCode/Notepad++).
- Create and save a file with .html extension.
- Write HTML code using formatting and list tags.
- Run the file in a browser.
- Observe the output.

5. Implementation with Code

Question 1: Basic structure of the webpage

- Use appropriate HTML tags to define the document structure
- Add a main heading displaying the title of the portal
- create a footer section containing the portal name and laboratory information.

Emerging Technologies in Information Technology
Emerging Technologies Information Portal Web Programming Laboratory

Program

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Emerging Technologies</title>

</head>

<body>

    <h1 align="center">Emerging Technologies in Information Technology</h1>

    <hr>

    <hr>

<footer align="center">

    <h2>Emerging Technologies Information Portal</h2>

    <p>Web Programming Laboratory</p>

</footer>

</body>

</html>
```

Question 2: text formatting tags

- Add an introduction section describing the importance of emerging technologies.
- Use different text formatting tags

Emerging Technologies in Information Technology

Introduction

Emerging technologies are transforming the way we live, work, and communicate. Technologies such as **Artificial Intelligence**, *Cloud Computing*, Cyber Security, and **Internet of Things (IoT)** are driving innovation across various industries.

Use of Text Formatting Tags

Bold Text: Artificial Intelligence enables machines to learn from data.

Italic Text: Cloud Computing provides scalable computing resources.

Underlined Text: Cyber Security protects digital assets.

Strong Text: Data is one of the most valuable resources today.

Emphasized Text: Emerging technologies are creating new career opportunities.

Water is represented as H₂O using subscript.

Einstein's famous equation is $E = mc^2$.

~~Traditional Storage Systems~~ are increasingly being replaced by cloud storage.

Artificial Intelligence is considered one of the most influential technologies of the decade.

Emerging Technologies Information Portal
Web Programming Laboratory

Program:

```
<!DOCTYPE html>

<html>

<head>

    <title>Emerging Technologies</title>

</head>

<body border="1">

    <center>

        <h1>Emerging Technologies in Information Technology</h1>

    </center>

    <hr>
```

<h2>Introduction</h2>

<p>

Emerging technologies are transforming the way we live, work, and communicate.

Technologies such as Artificial Intelligence,</p>

<i>Cloud Computing</i>,</p>

<u>Cyber Security</u>, and</p>

<mark>Internet of Things (IoT)</mark>

are driving innovation across various industries.

</p>

<h2>Use of Text Formatting Tags</h2>

<p>Bold Text: Artificial Intelligence enables machines to learn from data.</p>

<p><i>Italic Text:</i> Cloud Computing provides scalable computing resources.</p>

<p><u>Underlined Text:</u> Cyber Security protects digital assets.</p>

<p>Strong Text: Data is one of the most valuable resources today.</p>

<p>Emphasized Text: Emerging technologies are creating new career opportunities.</p>

<p>Water is represented as H₂O using subscript.</p>

<p>Einstein's famous equation is $E = mc^2$.</p>

<p>Traditional Storage Systems are increasingly being replaced by cloud storage.</p>

<p><mark>Artificial Intelligence</mark> is considered one of the most influential technologies of the decade.</p>

<hr>

<center>

<h3>Emerging Technologies Information Portal</h3>

<h3>Web Programming Laboratory</h3>

</center>

</body>

</html>

Question 3: Different Types of List

Top Emerging Technologies

1. Artificial Intelligence (AI)
2. Cloud Computing
3. Cyber Security
4. Internet of Things (IoT)
5. Blockchain

Benefits of Emerging Technologies

- Improves productivity
- Enhances decision making
- Provides better security
- Supports automation
- Creates new career opportunities

Technology Definitions

Artificial Intelligence (AI)

Technology that enables machines to simulate human intelligence and perform tasks such as learning and decision-making.

Cloud Computing

Delivery of computing services such as storage, servers, and applications over the Internet.

Cyber Security

Protection of computer systems, networks, and data from cyber threats and unauthorized access.

Internet of Things (IoT)

A network of interconnected devices that collect and exchange data through the Internet.

Blockchain

A distributed digital ledger used to securely record transactions across multiple systems.

Emerging Technologies Information Portal
Web Programming Laboratory

<!DOCTYPE html>

<html>

<head>

<title>Emerging Technologies</title>

</head>

<body>

<h1 align ="center">Emerging Technologies in Information Technology</h1>

<hr>

<h2>Introduction</h2>

<p>

Emerging technologies are transforming the way we live, work, and communicate.

Technologies such as Artificial Intelligence,</p>

<i>Cloud Computing</i>,</p>

<u>Cyber Security</u>, and</p>

<mark>Internet of Things (IoT)</mark>

are driving innovation across various industries.</p>

</p>

<h2>Use of Text Formatting Tags</h2>

<p>Bold Text: Artificial Intelligence enables machines to learn from data.</p>

<p><i>Italic Text:</i> Cloud Computing provides scalable computing resources.</p>

<p><u>Underlined Text:</u> Cyber Security protects digital assets.</p>

<p>Strong Text: Data is one of the most valuable resources today.</p>

<p>Emphasized Text: Emerging technologies are creating new career opportunities.</p>

<p>Water is represented as H₂O using subscript.</p>

<p>Einstein's famous equation is $E = mc^2$.</p>

<p>Traditional Storage Systems are increasingly being replaced by cloud storage.</p>

<p><mark>Artificial Intelligence</mark> is considered one of the most influential technologies of the decade.</p>

<h1>Top Emerging Technologies</h1>

Artificial Intelligence (AI)

Cloud Computing

Cyber Security

Internet of Things (IoT)

Blockchain

<h1>Benefits of Emerging Technologies</h1>

Improves productivity

Enhances decision making

Provides better security

Supports automation

Creates new career opportunities

<h1>Technology Definitions</h1>

<dl>

<dt>Artificial Intelligence (AI)</dt>

<dd>Technology that enables machines to simulate human intelligence and perform tasks such as learning and decision-making.</dd>

<dt>Cloud Computing</dt>

<dd>Delivery of computing services such as storage, servers, and applications over the Internet.</dd>

<dt>Cyber Security</dt>

<dd>Protection of computer systems, networks, and data from cyber threats and unauthorized access.</dd>

<dt>Internet of Things (IoT)</dt>

<dd>A network of interconnected devices that collect and exchange data through the Internet.</dd>

<dt>Blockchain</dt>

<dd>A distributed digital ledger used to securely record transactions across multiple systems.</dd>

</dl>

```
<hr>

<footer align="center">

    <h3>Emerging Technologies Information Portal</h3>

    <h3>Web Programming Laboratory</h3>

</footer>

</body>

</html>
```

6. Result

Successfully performed all the questions.

7. Applications

- Web page structure
- Content presentation

- Information hierarchy

8. Practice Questions

1. Nested List

Design an HTML page showing a nested list structure (at least 2 levels deep) representing the following hierarchy:

- Main topic: *Web Technologies* (ordered list)
- Sub-topics: HTML, CSS, JavaScript (each as ordered sub-items)
- Under HTML: Tags, Attributes, Structure (unordered sub-sub-items)
- Under CSS: Selectors, Properties, Box Model
- Under JavaScript: Variables, Functions, Events

Expected Output:

Web Technologies

1. HTML
 - Tags
 - Attributes
 - Structure
2. CSS
 - Selectors
 - Properties
 - Box Model
3. JavaScript
 - Variables
 - Functions
 - Events

2. Prepare the Scientific Report as shown below

Experiment: Electrolysis of Water

Decompose H_2O into H_2 and O_2 using D.C.

Energy: 1.23×10^3 mV

Observation: glowing splint relighted, confirming O_2 .

Command: `python log_data.py --voltage 12`

Answer

3. Create a web page for Blog Post as shown below

Why Learn HTML in 2025?

June 1, 2025 by **Riya Sharma**

HTML is the foundation of every website you have visited. Learning HTML is the best first step.

"In the beginner's mind there are many possibilities." — Shunryu Suzuki

Top reasons to learn HTML:

- Free, no special software
- Visible instantly in any browser
- Gateway to CSS and JavaScript
- Boosts employability

9. Questions (Viva / Oral)

1. What is HTML? What does it stand for?
2. What is the difference between `<b>` and `<strong>` tags?
3. What is the difference between `<i>` and `<em>` tags?
4. What are the six heading tags in HTML? Which is the largest and which is the smallest?
5. What is the use of `<sup>` and `<sub>` tags? Give one example each.
6. What is the difference between an ordered list and an unordered list?
7. Which tag is used to create a definition list? Name all three tags involved.
8. How do you change the bullet style in an unordered list?
9. What is a nested list? Give a real-life example.
10. What is the use of the `<mark>` tag?

PRACTICAL 2

1. Aim

Enhance the "Emerging Technologies Information Portal" by adding hyperlinks, semantic HTML tags, and a technology information table.

2. Prior Concepts

- Basic HTML structure
- Use of text formatting tags and lists
- Use of web browser and text editor

3. New Concepts

- Hyperlinks using <a> tag
- Internal, external and email hyperlinks
- Tables using <table>, <tr>, <th>, <td>
- Table attributes: border, cellpadding, cellspacing, rowspan, colspan
- Semantic tags: <header>, <nav>, <section>, <article>, <aside>, <footer>

4. Procedure

- Open a text editor (VSCode/Notepad++).
- Create and save a file with .html extension.
- Write HTML code using hyperlinks, tables and semantic tags.
- Run the file in a browser.
- Observe the output.

5. Implementation with Code

Question 1: Web Page Structure using Semantic Tags

- **Modify the webpage created in Q.1 by organizing the content using appropriate semantic HTML5 tags.**
- **Represent the page structure using Header, Navigation, Content Sections, Articles, and Footer.**
- **Add Header and Footer**

Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Emerging Technologies</title>
```

```
</head>
```

<body>

<!-- Header Section -->

<header>

<h1>Emerging Technologies in Information Technology</h1>

<p>Explore the latest technologies shaping the digital world.</p>

</header>

<hr>

<!-- Navigation Section -->

<nav>

Introduction |

Text Formatting |

Technologies |

Benefits |

Definitions

</nav>

<hr>

<!-- Main Content -->

<main>

<!-- Introduction Section -->

<section id="introduction">

<h2>Introduction</h2>

<article>

<p>

Emerging technologies are transforming the way we live, work,
and communicate. Technologies such as
Artificial Intelligence,
Cloud Computing,
<u>Cyber Security</u>, and
<mark>Internet of Things (IoT)</mark>
are driving innovation across various industries.

</p>

</article>

</section>

<!-- Text Formatting Section -->

<section id="formatting">

<h2>Use of Text Formatting Tags</h2>

<article>

<p>

Bold Text:

Artificial Intelligence enables machines to learn from data.

</p>

<p>

<i>Italic Text:</i>

Cloud Computing provides scalable computing resources.

</p>

<p>

<u>Underlined Text:</u>

Cyber Security protects digital assets.

</p>

<p>

Strong Text:

Data is one of the most valuable resources today.

</p>

<p>

Emphasized Text:

Emerging technologies are creating new career opportunities.

</p>

<p>

Water is represented as H₂O using subscript.

</p>

<p>

Einstein's famous equation is $E = mc^2$.

</p>

<p>

Traditional Storage Systems

are increasingly being replaced by cloud storage.

</p>

<p>

<mark>Artificial Intelligence</mark>

is considered one of the most influential technologies of the
decade.

</p>

</article>

</section>

<!-- Technologies Section -->

<section id="technologies">

<h2>Top Emerging Technologies</h2>

<article>

Artificial Intelligence (AI)

Cloud Computing

Cyber Security

Internet of Things (IoT)

Blockchain

</article>

</section>

<!-- Benefits Section -->

<section id="benefits">

<h2>Benefits of Emerging Technologies</h2>

```
<article>

  <ul>

    <li>Improves productivity</li>

    <li>Enhances decision-making</li>

    <li>Provides better security</li>

    <li>Supports automation</li>

    <li>Creates new career opportunities</li>

  </ul>

</article>

</section>
```

<!-- Definitions Section -->

```
<section id="definitions">

  <h2>Technology Definitions</h2>
```

```
<article>

  <dl>

    <dt><strong>Artificial Intelligence (AI)</strong></dt>

    <dd>

      Technology that enables machines to simulate human

      intelligence and perform tasks such as learning and

      decision-making.

    </dd>

    <dt><strong>Cloud Computing</strong></dt>

    <dd>

      Delivery of computing services such as storage, servers,

      and applications over the Internet.
```

</dd>

<dt>Cyber Security</dt>

<dd>

Protection of computer systems, networks, and data from
cyber threats and unauthorized access.

</dd>

<dt>Internet of Things (IoT)</dt>

<dd>

A network of interconnected devices that collect and
exchange data through the Internet.

</dd>

<dt>Blockchain</dt>

<dd>

A distributed digital ledger used to securely record
transactions across multiple systems.

</dd>

</dl>

</article>

</section>

</main>

<hr>

<!-- Footer Section -->

```
<footer>

    <h3>Emerging Technologies Information Portal</h3>

    <p>Web Programming Laboratory</p>

    <p>&copy; 2026 All Rights Reserved</p>

</footer>
```

```
</body>
```

```
</html>
```

Question 2: Internal Hyperlinks

- Create a navigation menu below the header. Add hyperlinks for the following sections: **Home , Artificial Intelligence, Cloud Computing, Cyber Security, Internet of Things, Technology Information, Contact Us**
- The hyperlinks should allow users to navigate to the corresponding sections within the same webpage.

Emerging Technologies in Information Technology

Exploring the technologies shaping the future

[Home](#) | [AI](#) | [Cloud Computing](#) | [Cyber Security](#) | [IoT](#) | [Technology Information](#) | [Contact Us](#)

Emerging Technologies Information Portal
Web Programming Laboratory

Code:

```
<!DOCTYPE html>

<html>

<head>

    <title>Emerging Technologies Information Portal</title>

</head>

<body>
```

<header align="center">

<h1>Emerging Technologies in Information Technology</h1>

<p>Exploring the technologies shaping the future</p>

</header>

<hr>

<nav align="center">

Home |

AI |

Cloud Computing |

Cyber Security |

IoT |

Technology Information |

Contact Us

</nav>

<hr>

<main>

<section id="home">

<h2>Introduction</h2>

<p>

Emerging technologies are transforming industries and creating new opportunities.

Technologies such as Artificial Intelligence, <i>Cloud Computing</i>,

<u>Cyber Security</u>, and <mark>Internet of Things (IoT)</mark> are driving

innovation worldwide.

</p>

</section>

<section id="ai">

<article>

<h2>Artificial Intelligence</h2>

<p>

Artificial Intelligence enables computers to perform tasks that normally
require human intelligence such as learning, reasoning, and decision-making.

</p>

</article>

</section>

<section id="cloud">

<article>

<h2>Cloud Computing</h2>

<p>

Cloud Computing provides on-demand access to computing resources such as
servers, storage, and applications over the internet.

</p>

</article>

</section>

<section id="cyber">

<article>

<h2>Cyber Security</h2>

<p>

Cyber Security protects computer systems, networks, and data from cyber

threats and unauthorized access.

</p>

</article>

</section>

<section id="iot">

<article>

<h2>Internet of Things (IoT)</h2>

<p>

IoT connects physical devices to the internet, enabling them to collect
and exchange data.

</p>

</article>

</section>

<section id="info">

<h2>Technology Information</h2>

<section id="contact">

<h2>Contact Us</h2>

<p>For more information about emerging technologies, feel free to contact us.</p>

<p>Organization: Emerging Technologies Information Portal</p>

<p>Email: info@techportal.com</p>

<p>Phone: +91 9876543210</p>

<p>Address: Tech Portal, Mumbai, Maharashtra, India</p>

</section>

</main>

<hr>

<footer align="center">

<p>Emerging Technologies Information Portal</p>

<p>Web Programming Laboratory</p>

</footer>

</body>

</html>

Question 3:

- Create separate sections for the technologies mentioned in the above question (AI, Cloud Computing, Cyber Security and IoT) and add a brief description for each.

Introduction

Emerging technologies are transforming industries and creating new opportunities. Technologies such as **Artificial Intelligence**, *Cloud Computing*, *Cyber Security*, and **Internet of Things (IoT)** are driving innovation worldwide.

Artificial Intelligence

Artificial Intelligence (AI) enables computers to perform tasks that normally require human intelligence such as learning, reasoning, and decision-making.

Cloud Computing

Cloud Computing provides on-demand access to computing resources such as servers, storage, and applications over the internet.

Cyber Security

Cyber Security protects computer systems, networks, and data from cyber threats and unauthorized access.

Internet of Things (IoT)

IoT connects physical devices to the internet, enabling them to collect and exchange data.

Code:

```
<!DOCTYPE html>

<html>

<head>

    <title>Emerging Technologies Information Portal</title>

</head>

<body>

<header align="center">

    <h1>Emerging Technologies in Information Technology</h1>

    <p>Exploring the technologies shaping the future</p>

</header>
```

<hr>

<nav align="center">

Home |

AI |

Cloud Computing |

Cyber Security |

IoT |

Technology Information |

Contact Us

</nav>

<hr>

<main>

<section id="home">

<h2>Introduction</h2>

<p>

Emerging technologies are transforming industries and creating new opportunities.

Technologies such as Artificial Intelligence, <i>Cloud Computing</i>,</p>

<u>Cyber Security</u>, and <mark>Internet of Things (IoT)</mark> are driving

innovation worldwide.

</p>

</section>

<section id="ai">

<article>

<h2>Artificial Intelligence</h2>

<p>

Artificial Intelligence enables computers to perform tasks that normally require human intelligence such as learning, reasoning, and decision-making.

</p>

</article>

</section>

<section id="cloud">

<article>

<h2>Cloud Computing</h2>

<p>

Cloud Computing provides on-demand access to computing resources such as servers, storage, and applications over the internet.

</p>

</article>

</section>

<section id="cyber">

<article>

<h2>Cyber Security</h2>

<p>

Cyber Security protects computer systems, networks, and data from cyber threats and unauthorized access.

</p>

</article>

</section>

```
<section id="iot">
```

```
<article>
```

```
<h2>Internet of Things (IoT)</h2>
```

```
<p>
```

```
IoT connects physical devices to the internet, enabling them to collect  
and exchange data.
```

```
</p>
```

```
</article>
```

```
</section>
```

```
<section id="info">
```

```
<h2>Technology Information</h2>
```

```
<table border="1" cellpadding="10" cellspacing="0">
```

```
<tr>
```

```
<th>Technology Name</th>
```

```
<th>Category</th>
```

```
<th>Year Introduced</th>
```

```
</tr>
```

```
<tr>
```

```
<td>Artificial Intelligence</td>
```

```
<td>Software Technology</td>
```

```
<td>1956</td>
```

```
</tr>
```

```
<tr>
```

```
<td>Cloud Computing</td>
```

```
<td>Infrastructure Technology</td>
```

	<td>2006</td>
	</tr>
	<tr>
	<td>Cyber Security</td>
	<td>Security Technology</td>
	<td>1990</td>
	</tr>
	<tr>
	<td>Internet of Things</td>
	<td>Networking Technology</td>
	<td>1999</td>
	</tr>
	<tr>
	<td>Blockchain</td>
	<td>Distributed Technology</td>
	<td>2008</td>
	</tr>
	</table>
	</section>

<section id="contact">

<h2>Contact Us</h2>

<p>For more information about emerging technologies, feel free to contact us.</p>

<p>Organization: Emerging Technologies Information Portal</p>

<p>Email: info@techportal.com</p>

<p>Phone: +91 9876543210</p>

<p>Address: Tech Portal, Mumbai, Maharashtra, India</p>

```
</section>
```

```
</main>
```

```
<hr>
```

```
<footer align="center">
```

```
  <p><b>Emerging Technologies Information Portal</b></p>
```

```
  <p>Web Programming Laboratory</p>
```

```
</footer>
```

```
</body>
```

```
</html>
```

Question 4: Basic Table

- Create a table displaying information about emerging technologies. The table should contain the following details:

Technology Name	Category	Year Introduced
Artificial Intelligence	Software Technology	1956
Cloud Computing	Infrastructure Technology	2006
Cyber Security	Security Technology	1990
Internet of Things	Networking Technology	1999
Blockchain	Distributed Technology	2008

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Emerging Technologies Information Portal</title>

</head>

<body>

<header align="center">

    <h1>Emerging Technologies in Information Technology</h1>

    <p>Exploring the technologies shaping the future</p>

</header>

<hr>

<nav align="center">

    <a href="#home">Home</a> |

    <a href="#ai">AI</a> |

    <a href="#cloud">Cloud Computing</a> |

    <a href="#cyber">Cyber Security</a> |

    <a href="#iot">IoT</a> |

    <a href="#info">Technology Information</a> |

    <a href="#contact">Contact Us</a>

</nav>

<hr>

<main>

<section id="home">

    <h2>Introduction</h2>

    <p>
```

Emerging technologies are transforming industries and creating new opportunities.

Technologies such as **Artificial Intelligence**, *Cloud Computing*,
Cyber Security, and **Internet of Things (IoT)** are driving
innovation worldwide.

Artificial Intelligence

Artificial Intelligence enables computers to perform tasks that normally
require human intelligence such as learning, reasoning, and decision-making.

Cloud Computing

Cloud Computing provides on-demand access to computing resources such as
servers, storage, and applications over the internet.

<article>

<h2>Cyber Security</h2>

<p>

Cyber Security protects computer systems, networks, and data from cyber threats and unauthorized access.

</p>

</article>

</section>

<section id="iot">

<article>

<h2>Internet of Things (IoT)</h2>

<p>

IoT connects physical devices to the internet, enabling them to collect and exchange data.

</p>

</article>

</section>

<section id="info">

<h2>Technology Information</h2>

<table border="1" cellpadding="10" cellspacing="0">

<tr>

<th>Technology Name</th>

<th>Category</th>

<th>Year Introduced</th>

</tr>

<tr>

<td>Artificial Intelligence</td>

<td>Software Technology</td>

<td>1956</td>

</tr>

<tr>

<td>Cloud Computing</td>

<td>Infrastructure Technology</td>

<td>2006</td>

</tr>

<tr>

<td>Cyber Security</td>

<td>Security Technology</td>

<td>1990</td>

</tr>

<tr>

<td>Internet of Things</td>

<td>Networking Technology</td>

<td>1999</td>

</tr>

<tr>

<td>Blockchain</td>

<td>Distributed Technology</td>

<td>2008</td>

</tr>

</table>

</section>

```
<section id="contact">
```

```
    <h2>Contact Us</h2>
```

```
</section>
```

```
</main>
```

```
<hr>
```

```
<footer align="center">
```

```
    <p><b>Emerging Technologies Information Portal</b></p>
```

```
    <p>Web Programming Laboratory</p>
```

```
</footer>
```

```
</body>
```

```
</html>
```

Question 5: Contact Us Section

- Create a **Contact Us** section at the bottom of the webpage.
- Display the organization's contact information and use appropriate hyperlinks for email and telephone communication.

The section should contain:

- Organization Name
- Email Address
- Contact Number
- Office Address

Contact Us

For more information about emerging technologies, feel free to contact us.

Organization: Emerging Technologies Information Portal

Email: info@techportal.com

Phone: [+91 9876543210](tel:+919876543210)

Address: Tech Portal, Mumbai, Maharashtra, India

Code:

```
<!DOCTYPE html>

<html>

<head>

    <title>Emerging Technologies Information Portal</title>

</head>

<body>


<header align="center">

    <h1>Emerging Technologies in Information Technology</h1>

    <p>Exploring the technologies shaping the future</p>

</header>


<hr>


<nav align="center">

    <a href="#home">Home</a> |

    <a href="#ai">AI</a> |

    <a href="#cloud">Cloud Computing</a> |

    <a href="#cyber">Cyber Security</a> |

    <a href="#iot">IoT</a> |

    <a href="#info">Technology Information</a> |

    <a href="#contact">Contact Us</a>

</nav>


<hr>


<main>
```

<section id="home">

<h2>Introduction</h2>

<p>

Emerging technologies are transforming industries and creating new opportunities.

Technologies such as Artificial Intelligence, <i>Cloud Computing</i>,</p>

<u>Cyber Security</u>, and <mark>Internet of Things (IoT)</mark> are driving

innovation worldwide.

</p>

</section>

<section id="ai">

<article>

<h2>Artificial Intelligence</h2>

<p>

Artificial Intelligence enables computers to perform tasks that normally

require human intelligence such as learning, reasoning, and decision-making.

</p>

</article>

</section>

<section id="cloud">

<article>

<h2>Cloud Computing</h2>

<p>

Cloud Computing provides on-demand access to computing resources such as

servers, storage, and applications over the internet.

</p>

</article>

</section>

<section id="cyber">

<article>

<h2>Cyber Security</h2>

<p>

Cyber Security protects computer systems, networks, and data from cyber threats and unauthorized access.

</p>

</article>

</section>

<section id="iot">

<article>

<h2>Internet of Things (IoT)</h2>

<p>

IoT connects physical devices to the internet, enabling them to collect and exchange data.

</p>

</article>

</section>

<section id="info">

<h2>Technology Information</h2>

<table border="1" cellpadding="10" cellspacing="0">

<tr>

<th>Technology Name</th>
<th>Category</th>
<th>Year Introduced</th>
</td>
<td>Artificial Intelligence</td>
<td>Software Technology</td>
<td>1956</td>
<td>Cloud Computing</td>
<td>Infrastructure Technology</td>
<td>2006</td>
<td>Cyber Security</td>
<td>Security Technology</td>
<td>1990</td>
<td>Internet of Things</td>
<td>Networking Technology</td>
<td>1999</td>
<td>Blockchain</td>
<td>Distributed Technology</td>
<td>2008</td>

```
</tr>

</table>

</section>

<section id="contact">

  <h2>Contact Us</h2>

  <p>For more information about emerging technologies, feel free to contact us.</p>

  <p><b>Organization:</b> Emerging Technologies Information Portal</p>

  <p><b>Email:</b> <a href="mailto:info@techportal.com">info@techportal.com</a></p>

  <p><b>Phone:</b> <a href="tel:+919876543210">+91 9876543210</a></p>

  <p><b>Address:</b> Tech Portal, Mumbai, Maharashtra, India</p>

</section>

</main>

<hr>

<footer align="center">

  <p><b>Emerging Technologies Information Portal</b></p>

  <p>Web Programming Laboratory</p>

</footer>

</body>

</html>
```

6. Result

Successfully performed all the questions.

7. Applications

- Website navigation
- Data representation using tables
- Semantic and accessible webpage design

8. Practice Questions

1. Write an HTML program to create the following timetable using appropriate semantic tags and table attributes.

Examination Timetable

TYIT Mid-Semester Examination Schedule

Day	Time	Subject
Monday	10:00 AM - 12:00 PM	Web Programming
	1:00 PM - 3:00 PM	Python Programming
Tuesday	10:00 AM - 12:00 PM	Database Systems
	1:00 PM - 3:00 PM	Data Analytics

Department of Information Technology

Answer

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

2. Department Faculty Information

Write an HTML program to create the following webpage.

Department of Computing

About Department

The Department of Computing offers undergraduate and postgraduate programmes that prepare students for careers in software development, networking, cybersecurity, data science, and emerging technologies.

Faculty Information

Name	Designation
Dr. Asif Rampurawala	Head of Department
Dr. Pushpa Mahapatro	Programme Coordinator

Quick Links

- [Courses](#)
- [Placements](#)
- [Contact](#)

Contact Information

Department of Computing
VSIT, Mumbai
Email: itdept@xyzcollege.edu
Phone: [+91 9876543210](tel:+919876543210)

Answer

9. Questions (Viva / Oral)

1. What is a hyperlink in HTML?
2. What is the difference between internal and external hyperlinks?
3. What is the purpose of target="_blank"?
4. Which tag is used to create tables?
5. What is the difference between <th> and <td>?
6. What is rowspan?
7. What is colspan?
8. What are semantic tags?
9. Name any four semantic tags in HTML5.
10. What are the advantages of semantic HTML?

PRACTICAL 3

1. Aim

Enhance the "Emerging Technologies Information Portal" by applying CSS styles and displaying graphics related to emerging technologies.

2. Prior Concepts

- Basic HTML structure
- Text formatting tags
- Hyperlinks and tables

3. New Concepts

- CSS syntax and selectors
- Inline, Internal and External CSS
- Text and paragraph formatting properties
- Image insertion using tag
- Background images and styling

4. Procedure

- Open a text editor (VSCode/Notepad++).
- Create and save HTML and CSS files.
- Apply different styles to text and paragraphs.

- Insert graphics/images into the webpage.
- Run the file in a browser and observe the output.

5. Implementation with Code

Question 1:

Extend the "Emerging Technologies Information Portal" by adding suitable images for the following technologies:

- Artificial Intelligence
- Cloud Computing
- Cyber Security
- Internet of Things

Display the image, technology name, and a brief description for each technology.

Emerging Technologies Information Portal

Exploring Technologies Shaping the Future

[Home](#) | [AI](#) | [Cloud Computing](#) | [Cyber Security](#) | [IoT](#) | [Technology Information](#) | [Contact Us](#)

Introduction

Emerging technologies are transforming industries and creating innovative solutions for businesses and society. Technologies such as Artificial Intelligence, Cloud Computing, Cyber Security, and Internet of Things are driving digital transformation across the world.

Artificial Intelligence (AI)



Artificial Intelligence enables machines to perform tasks that normally require human intelligence such as learning, problem-solving, and decision-making.

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Emerging Technologies Information Portal</title>

</head>

<body>

  <!-- Header -->

  <header>

    <h1>Emerging Technologies Information Portal</h1>

    <p>Exploring Technologies Shaping the Future</p>

    <hr>

    <!-- Navigation -->

    <nav>

      <a href="#ai">AI</a> |

      <a href="#cloud">Cloud Computing</a> |

      <a href="#cyber">Cyber Security</a> |

      <a href="#iot">IoT</a> |

      <a href="#contact">Contact Us</a>

    </nav>

    <hr>

  </header>

  <!-- Introduction -->

  <section>

    <h2>Introduction</h2>

    <p>

      Emerging technologies are transforming industries by providing

      innovative solutions. Artificial Intelligence, Cloud Computing,

      Cyber Security, and Internet of Things (IoT) are among the most

      important technologies today.

    </p>

  </section>

</body>
```

</p>

</section>

<hr>

<!-- Artificial Intelligence -->

<article id="ai">

<h2>Artificial Intelligence (AI)</h2>

<p>

Artificial Intelligence enables computers to perform tasks such as
learning, reasoning, problem-solving, and decision-making.

</p>

</article>

<hr>

<!-- Cloud Computing -->

<article id="cloud">

<h2>Cloud Computing</h2>

<p>

Cloud Computing provides on-demand access to computing resources
such as servers, storage, and applications over the Internet.

</p>

</article>

<hr>

<!-- Cyber Security -->

<article id="cyber">

<h2>Cyber Security</h2>

<p>

Cyber Security protects computer systems, networks, and data from
cyber threats, hacking, and unauthorized access.

</p>

</article>

<hr>

<!-- Internet of Things -->

<article id="iot">

<h2>Internet of Things (IoT)</h2>

<p>

The Internet of Things (IoT) connects smart devices through the
Internet, enabling them to communicate and share data.

</p>

</article>

<hr>

<!-- Footer -->

<footer id="contact">

<p>Contact Us: info@technologyportal.com</p>

<p>© 2026 Emerging Technologies Information Portal</p>

</footer>

</body>

</html>

Question 2:

Organize the technology information into a card-based layout. Each card should contain:

- Technology Image
- Technology Name
- Brief Description

Arrange all technology cards within the same webpage.

Code:

<!DOCTYPE html>

<html>

<head>

<title>Technology Cards</title>

</head>

<body>

<h1 align="center">Emerging Technologies</h1>

```
<table align="center" cellpadding="20">
```

```
<tr>
```

```
<!-- AI Card -->
```

```
<td align="center" border="1">
```

```
<br><br>
```

```
<h3>Artificial Intelligence</h3>
```

```
<p>
```

```
    AI enables machines to learn,  
    solve problems, and make intelligent decisions.
```

```
</p>
```

```
</td>
```

```
<!-- Cloud Card -->
```

```
<td align="center">
```

```
<br><br>
```

```
<h3>Cloud Computing</h3>
```

```
<p>
```

```
    Cloud Computing provides online  
    storage, servers, and software services.
```

```
</p>
```

```
</td>
```

```
</tr>
```

```
<tr>
```

```
<!-- Cyber Security Card -->
```

```
<td align="center">
```

```
<br><br>
```

<h3>Cyber Security</h3>

<p>

Cyber Security protects systems,
networks, and data from cyber attacks.

</p>

</td>

<!-- IoT Card -->

<td align="center">

<h3>Internet of Things</h3>

<p>

IoT connects smart devices over
the Internet for communication and automation.

</p>

</td>

</tr>

</table>

</body>

</html>

Question 3:

Apply Inline CSS to improve the appearance of the technology cards.

Format:

- Card background color

- Text color
- Border
- Alignment

[Home](#) | [Technologies](#) | [Technology Information](#) | [Contact Us](#)

Introduction

Emerging technologies are transforming industries and creating innovative solutions for businesses and society. Technologies such as Artificial Intelligence, Cloud Computing, Cyber Security, and Internet of Things are driving digital transformation across the world.

Emerging Technologies

Artificial Intelligence (AI)



Artificial Intelligence enables machines to perform tasks that normally require human intelligence such as learning, reasoning, and decision-making.

Code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Technology Cards with Inline CSS</title>

</head>

<body style="font-family:Arial; background-color:#f5f5f5;">

  <h1 style="text-align:center; color:darkblue;">
```

Emerging Technologies

</h1>

<table align="center" cellpadding="20">

<tr>

<!-- AI Card -->

<td style="background-color:#E6F2FF;

border:2px solid blue;

text-align:center;

color:#003366;

width:250px;">

<h2>Artificial Intelligence</h2>

<p>

AI enables machines to learn,
reason, and make intelligent decisions.

</p>

</td>

<!-- Cloud Card -->

<td style="background-color:#E8F8F5;

border:2px solid green;

text-align:center;

color:darkgreen;

width:250px;">

<h2>Cloud Computing</h2>

<p>

Cloud Computing provides online
storage and computing services.

</p>

</td>

</tr>

<tr>

<!-- Cyber Security Card -->

<td style="background-color:#FFF4E6;
border:2px solid orange;
text-align:center;
color:brown;
width:250px;">

<h2>Cyber Security</h2>

<p>

Cyber Security protects systems,

networks, and digital information.

</p>

</td>

<!-- IoT Card -->

<td style="background-color:#F4ECF7;

border:2px solid purple;

text-align:center;

color:purple;

width:250px;">

<h2>Internet of Things (IoT)</h2>

<p>

IoT connects smart devices to
communicate and exchange data.

</p>

</td>

</tr>

</table>

</body>

</html>

Question 4:

Replace the inline styles with an Internal Style Sheet and apply styles to:

- Technology Cards
- Images
- Headings
- Paragraphs

Code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Technology Cards - Internal CSS</title>

<style>

  body{

    font-family: Arial, sans-serif;

    background-color:#f2f2f2;

  }

  h1{

    text-align:center;

    color:darkblue;

  }

  table{

    margin:auto;
```

```
}
```

```
/* Technology Cards */
```

```
.card{
```

```
    background-color:#E6F2FF;
```

```
    border:2px solid blue;
```

```
    text-align:center;
```

```
    width:250px;
```

```
    padding:15px;
```

```
}
```

```
/* Images */
```

```
img{
```

```
    width:180px;
```

```
    height:120px;
```

```
    border:2px solid gray;
```

```
}
```

```
/* Headings */
```

```
h2{
```

```
    color:darkblue;
```

```
}
```

```
/* Paragraphs */
```

```
p{
```

```
    color:#333333;
```

```
    font-size:16px;
```

```
    text-align:justify;
```

```
}
```

```
</style>
```

</head>

<body>

<h1>Emerging Technologies</h1>

<table cellpadding="20">

<tr>

<td class="card">

<h2>Artificial Intelligence</h2>

<p>

AI enables machines to learn, reason, solve problems,
and make intelligent decisions.

</p>

</td>

<td class="card">

<h2>Cloud Computing</h2>

<p>

Cloud Computing provides online storage,
servers, and software services.

</p>

</td>

</tr>

```
<tr>
```

```
<td class="card">
```

```

```

```
<h2>Cyber Security</h2>
```

```
<p>
```

```
    Cyber Security protects computer systems,  
    networks, and digital information.
```

```
</p>
```

```
</td>
```

```
<td class="card">
```

```

```

```
<h2>Internet of Things (IoT)</h2>
```

```
<p>
```

```
    IoT connects smart devices through the  
    Internet for communication and automation.
```

```
</p>
```

```
</td>
```

```
</tr>
```

```
</table>
```

```
</body>
```

```
</html>
```

Question 5:

Create an External Style Sheet (style.css) and link it to the webpage.

Move all styles from the internal style sheet to the external CSS file.

Apply styles to:

- Header
- Navigation Menu
- Technology Cards
- Table
- Contact Section
- Footer

Emerging Technologies in Information Technology

Exploring the Technologies Shaping the Future

[Home](#)[AI](#)[Cloud Computing](#)[Cyber Security](#)[IoT](#)[Technology Table](#)

Introduction

Emerging technologies are transforming industries and creating innovative solutions for businesses and society. Artificial Intelligence, Cloud Computing, Cyber Security, and Internet of Things are among the most impactful technologies today.



Artificial Intelligence

AI enables computers to simulate human intelligence and perform tasks such as learning, reasoning, and decision-making.



Cloud Computing

Cloud computing provides scalable computing resources and storage over the internet.



Cyber Security

Cyber Security protects systems, networks, and data from cyber threats and unauthorized access.



Internet of Things (IoT)

IoT connects devices through the internet, allowing them to collect and exchange data.

Technology Information Table

Technology	Category	Year Introduced
Artificial Intelligence	Software Technology	1956
Cloud Computing	Infrastructure Technology	2006
Cyber Security	Security Technology	1990
Internet of Things	Networking Technology	1999

Emerging Technologies Information Portal | Web Programming Laboratory

Code:

1. HTML File (index.html)

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Emerging Technologies Information Portal</title>
```

```
    <!-- External CSS -->
```

```
    <link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<!-- Header -->
```

```
<header>
```

```
    <h1>Emerging Technologies Information Portal</h1>
```

```
    <p>Exploring Technologies Shaping the Future</p>
```

```
</header>
```

```
<!-- Navigation -->
```

```
<nav>
```

```
    <a href="#">Home</a>
```

```
    <a href="#">Technologies</a>
```

[Technology Information](#)

[Contact Us](#)

<!-- Introduction -->

<section>

<h2>Introduction</h2>

<p>

Emerging technologies are transforming industries and creating innovative solutions for businesses and society. Artificial Intelligence, Cloud Computing, Cyber Security, and Internet of Things are among the most impactful technologies today.

</p>

</section>

<h2 align="center">Emerging Technologies</h2>

<!-- Technology Cards -->

<table>

<tr>

<td>

<div class="card">

<h3>Artificial Intelligence</h3>

<p>

AI enables computers to simulate human intelligence and perform tasks such as learning, reasoning and decision-making.

</p>

</div>

</td>

<td>

<div class="card">

<h3>Cloud Computing</h3>

<p>

Cloud computing provides scalable computing resources and storage over the Internet.

</p>

</div>

</td>

<td>

<div class="card">

<h3>Cyber Security</h3>

<p>

Cyber Security protects systems, networks and data from cyber threats and unauthorized access.

</p>

</div>

</td>

<td>

<div class="card">

<h3>Internet of Things (IoT)</h3>

<p>

IoT connects devices through the Internet allowing them to collect and exchange data.

</p>

</div>

</td>

</tr>

</table>

<!-- Contact Section -->

<section class="contact">

<h2>Contact Us</h2>

<p>Email : info@technologyportal.com</p>

<p>Phone : +91-9876543210</p>

</section>

<!-- Footer -->

<footer>

<p>© 2026 Emerging Technologies Information Portal</p>

</footer>

</body>

</html>

File 2: style.css

/* Body */

body{

```
font-family: Arial, sans-serif;

background:#f2f2f2;

margin:20px;

}
```

```
/* Header */
```

```
header{

background:#003366;

color:white;

text-align:center;

padding:15px;

}
```

```
/* Navigation Menu */
```

```
nav{

background:#0059b3;

text-align:center;

padding:10px;

}
```

```
nav a{

color:white;

text-decoration:none;

margin:15px;

font-weight:bold;

}
```

```
nav a:hover{
```

```
    color:yellow;
}
```

```
/* Table */
```

```
table{
    margin:auto;
    border-spacing:20px;
}
```

```
/* Technology Cards */
```

```
.card{
    width:250px;
    background:white;
    border:2px solid #cccccc;
    border-radius:10px;
    text-align:center;
    padding:15px;
    box-shadow:3px 3px 8px gray;
}
```

```
.card img{
    width:220px;
    height:150px;
    border-radius:5px;
}
```

```
.card h3{
    color:#003366;
```

```
}
```

```
.card p{  
    text-align:justify;  
}
```

```
/* Contact Section */
```

```
.contact{  
    background:#e6f2ff;  
    border:2px solid #003366;  
    padding:15px;  
    margin-top:20px;  
}
```

```
/* Footer */
```

```
footer{  
    background:#003366;  
    color:white;  
    text-align:center;  
    padding:10px;  
    margin-top:20px;  
}
```

6. Result

Successfully performed all the questions.

7. Applications

- Website designing
- Visual presentation of content
- User interface enhancement

8. Practice Questions

1. The following HTML code creates a technology card. Apply suitable Internal CSS to:

- Set a background color
- Add a border
- Center align the content
- Add padding
- Round the card corners

```
<div class="card">

  <h3>Artificial Intelligence</h3>

  <p>

    AI enables machines to simulate human intelligence.

  </p>

</div>
```

Code:

[illegible]

2. Convert Inline CSS to Internal CSS

```
<h1 style="color:blue; text-align:center;">
    Emerging Technologies
</h1>
```

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

- Enhance the appearance of the Practical 3 webpage by applying additional CSS properties to achieve a more professional look.

Perform the following enhancements:

- Add hover effects to navigation links.
- Increase spacing using padding and margins.
- Apply box-shadow effects to technology cards.
- Improve table appearance by styling table headers and cells.
- Highlight the Contact Us section using background colors and borders.
- Add hover effects to technology cards.

9. Questions (Viva / Oral)

1. What is CSS?
2. What are the types of CSS?
3. Difference between inline and internal CSS?
4. What is an external style sheet?
5. What is a CSS selector?
6. What is the use of the `<img>` tag?
7. What is alternate text?

8. What is a background image?
9. What is text-decoration?
10. What are the advantages of CSS?

PRACTICAL 4

1. Aim

HTML Form Designing Using Different Controls

2. Prior Concepts

- Basic HTML tags
- CSS formatting
- Hyperlinks and tables

3. New Concepts

- HTML forms
- Input controls
- Form attributes
- Validation attributes

4. Procedure

- Create an HTML form.
- Add different input controls.
- Apply validation attributes.
- Execute and verify the output.

5. Implementation with Code

Question 1:

Create a webpage named **suggest.html** containing a form to collect information about an emerging technology.

Include:

- Technology Name
- Organization Name
- Description
- Submit Button
- Reset Button

Emerging Technologies in Information Technology

Exploring the Technologies Shaping the Future

[Home](#) | [Suggest Technology](#)

Suggest a Technology

Use the following form to suggest an emerging technology that can be added to the portal.

Technology Name:

Organization Name:

Description:

Emerging Technologies Information Portal | Web Programming Laboratory

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Suggest a Technology</title>
```

```
</head>
```

```
<body>
```

```
<!-- Header -->
```

```
<header>
```

<h1>Emerging Technologies in Information Technology</h1>

<p>Exploring the Technologies Shaping the Future</p>

<hr>

<!-- Navigation -->

<nav>

Home |

Suggest Technology

</nav>

<hr>

</header>

<!-- Suggest Technology Form -->

<section>

<h2>Suggest a Technology</h2>

<p>

Use the following form to suggest an emerging technology

that can be added to the portal.

</p>

<form>

<label>Technology Name:</label>

<input type="text" name="technology" size="30">

<label>Organization Name:</label>

<input type="text" name="organization" size="30">

<label>Description:</label>


```
<textarea name="description" rows="6" cols="40"></textarea><br><br>
```

```
<input type="submit" value="Submit">
```

```
<input type="reset" value="Reset">
```

```
</form>
```

```
</section>
```

```
<hr>
```

```
<!-- Footer -->
```

```
<footer>
```

```
<p>
```

```
    Emerging Technologies Information Portal |
```

```
    Web Programming Laboratory
```

```
</p>
```

```
</footer>
```

```
</body>
```

```
</html>
```

Question 2:

Modify the form to include a **Technology Category** dropdown list.

Options:

- Artificial Intelligence
- Cloud Computing
- Cyber Security
- IoT

- Blockchain

Suggest a Technology

Use the following form to suggest an emerging technology that can be added to the portal.

Technology Name:

Organization Name:

Technology Category:

Description:

Submit

Reset

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Suggest a Technology</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Suggest a Technology</h1>
```

```
  <p>
```

```
    Use the following form to suggest an emerging technology
```

```
    that can be added to the portal.
```

```
</p>
```

```
<form>
```

<!-- Technology Name -->

<label>Technology Name:</label>

<input type="text" name="technology">

<!-- Organization Name -->

<label>Organization Name:</label>

<input type="text" name="organization">

<!-- Technology Category -->

<label>Technology Category:</label>

<select name="category">

<option>Artificial Intelligence</option>

<option>Cloud Computing</option>

<option>Cyber Security</option>

<option>IoT</option>

<option>Blockchain</option>

</select>

<!-- Description -->

<label>Description:</label>

<textarea name="description" rows="6" cols="40"></textarea>

<!-- Buttons -->

<input type="submit" value="Submit">

<input type="reset" value="Reset">

</form>

</body>

</html>

Question 3: Add radio buttons to select the **Technology Impact Level**.

Options:

- High
- Medium
- Low

Suggest a Technology

Use the following form to suggest an emerging technology that can be added to the portal.

Technology Name:

Organization Name:

Technology Category:

Technology Impact Level:

☐ High
☐ Medium
☐ Low

Description:

Code:

<!DOCTYPE html>

<html>

<head>

<title>Suggest a Technology</title>

</head>

```
<body>
```

```
<h1>Suggest a Technology</h1>
```

```
<p>
```

```
    Use the following form to suggest an emerging technology  
    that can be added to the portal.
```

```
</p>
```

```
<form>
```

```
    <!-- Technology Name -->
```

```
    <label>Technology Name:</label><br>
```

```
    <input type="text" name="technology"><br><br>
```

```
    <!-- Organization Name -->
```

```
    <label>Organization Name:</label><br>
```

```
    <input type="text" name="organization"><br><br>
```

```
    <!-- Technology Category -->
```

```
    <label>Technology Category:</label><br>
```

```
    <select name="category">
```

```
        <option>-- Select Category --</option>
```

```
        <option>Artificial Intelligence</option>
```

```
        <option>Cloud Computing</option>
```

```
        <option>Cyber Security</option>
```

```
        <option>IoT</option>
```

```
        <option>Blockchain</option>
```

```
</select>
```


<!-- Technology Impact Level -->

<label>Technology Impact Level:</label>

<input type="radio" name="impact" value="High">

<label>High</label>

<input type="radio" name="impact" value="Medium">

<label>Medium</label>

<input type="radio" name="impact" value="Low">

<label>Low</label>

<!-- Description -->

<label>Description:</label>

<textarea name="description" rows="6" cols="40"></textarea>

<!-- Buttons -->

<input type="submit" value="Submit">

<input type="reset" value="Reset">

</form>

</body>

</html>

Question 4: Add checkboxes to select **Application Areas**.

Options:

- Healthcare
- Education
- Finance
- Manufacturing
- Transportation

Technology Category:

-- Select Category -- ▾

Technology Impact Level:

☐ High

☐ Medium

☐ Low

Application Areas:

☐ Healthcare

☐ Education

☐ Finance

☐ Manufacturing

☐ Transportation

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Suggest a Technology</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Suggest a Technology</h1>
```

<p>

Use the following form to suggest an emerging technology
that can be added to the portal.

</p>

<form>

<!-- Technology Name -->

<label>Technology Name:</label>

<input type="text" name="technology">

<!-- Organization Name -->

<label>Organization Name:</label>

<input type="text" name="organization">

<!-- Technology Category -->

<label>Technology Category:</label>

<select name="category">

<option>-- Select Category --</option>

<option>Artificial Intelligence</option>

<option>Cloud Computing</option>

<option>Cyber Security</option>

<option>IoT</option>

<option>Blockchain</option>

</select>

<!-- Technology Impact Level -->

<label>Technology Impact Level:</label>

<input type="radio" name="impact" value="High"> High

<input type="radio" name="impact" value="Medium"> Medium

<input type="radio" name="impact" value="Low"> Low

<!-- Application Areas -->

<label>Application Areas:</label>

<input type="checkbox" name="application" value="Healthcare"> Healthcare

<input type="checkbox" name="application" value="Education"> Education

<input type="checkbox" name="application" value="Finance"> Finance

<input type="checkbox" name="application" value="Manufacturing"> Manufacturing

<input type="checkbox" name="application" value="Transportation"> Transportation

<!-- Description -->

<label>Description:</label>

<textarea name="description" rows="6" cols="40"></textarea>

<!-- Buttons -->

<input type="submit" value="Submit">

```
<input type="reset" value="Reset">
```

```
</form>
```

```
</body>
```

```
</html>
```

Question 5: Enhance the form by adding HTML5 controls:

- Email
- Contact Number
- Website URL
- Date of Submission
- Year Introduced

Emerging Technologies Information Portal

Suggest a New Technology

[Home](#)

[Suggest Technology](#)

Technology Suggestion Form

Name:

Email:

Contact Number:

Technology Name:

Technology Category:

Select Category



Technology Type:

☐ Software ☐ Hardware

Application Areas:

☐ Education ☐ Healthcare ☐ Finance ☐ Industry

Year Introduced:

Technology Website:

Date of Submission:

dd-mm-yyyy



Description:

Submit

Reset

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Technology Suggestion Form</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Technology Suggestion Form</h1>
```

```
  <form>
```

```
    <!-- Name -->
```

```
    <label><b>Name:</b></label><br>
```

```
    <input type="text" name="name"><br><br>
```

```
    <!-- Email -->
```

```
    <label><b>Email:</b></label><br>
```

```
    <input type="email" name="email"><br><br>
```

```
    <!-- Contact Number -->
```

```
    <label><b>Contact Number:</b></label><br>
```

```
    <input type="tel" name="contact"><br><br>
```

```
    <!-- Technology Name -->
```

```
<label><b>Technology Name:</b></label><br>
<input type="text" name="technology"><br><br>
```

```
<!-- Technology Category -->
<label><b>Technology Category:</b></label><br>
<select name="category">
  <option>Select Category</option>
  <option>Artificial Intelligence</option>
  <option>Cloud Computing</option>
  <option>Cyber Security</option>
  <option>Internet of Things (IoT)</option>
  <option>Blockchain</option>
</select><br><br>
```

```
<!-- Website URL -->
<label><b>Website URL:</b></label><br>
<input type="url" name="website"><br><br>
```

```
<!-- Date of Submission -->
<label><b>Date of Submission:</b></label><br>
<input type="date" name="submissiondate"><br><br>
```

```
<!-- Year Introduced -->
<label><b>Year Introduced:</b></label><br>
<input type="number" name="year" min="1900" max="2100"><br><br>
```

```
<!-- Description -->
<label><b>Description:</b></label><br>
<textarea name="description" rows="5" cols="40"></textarea><br><br>
```

```
<!-- Buttons -->  
  
<input type="submit" value="Submit">  
  
<input type="reset" value="Reset">  
  
</form>
```

```
</body>
```

```
</html>
```

6. Result

Successfully performed all the questions.

7. Applications

- Online registration systems
- Feedback systems
- Data collection applications

8. Practice Questions

1. Modify the Technology Suggestion Form by adding:

- Color Picker
- Range Slider
- File Upload Control

Use appropriate HTML5 input types.

Code:

2. Online Book Store Registration Form - Create a form to collect the following information:

- Customer Name
- Email Address
- Mobile Number

- Preferred Book Category (Dropdown)
- Membership Type (Radio Buttons)
- Interested Genres (Checkboxes)
- Suggestions (Text Area)
- Submit and Reset Buttons
- Design an Event Registration Form.

3. Create a Gym Membership Form.

Include:

- Member Name
- Age
- Gender
- Membership Plan (Dropdown)
- Fitness Goals (Checkboxes)
- Contact Number
- Email
- Joining Date

9. Questions (Viva / Oral)

1. What is an HTML form?
2. What is the action attribute?
3. What is the method attribute?
4. Difference between GET and POST?
5. What is a textbox?
6. What is a textarea?
7. Difference between radio button and checkbox?
8. What is a dropdown list?
9. What is the required attribute?
10. What is form validation?

PRACTICAL 5

1. Aim

2. Prior Concepts

- HTML forms
- Input controls
- Basic programming concepts

3. New Concepts

- JavaScript syntax
- Variables and operators
- Functions
- Events and validation

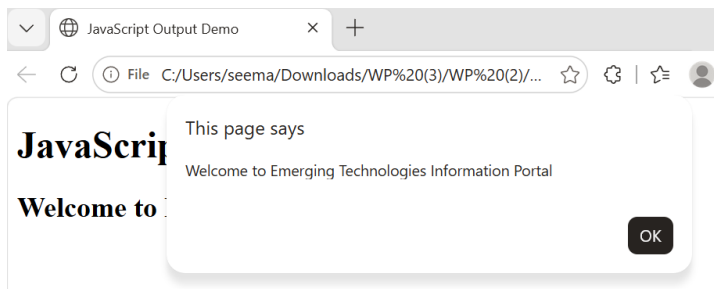
4. Procedure

- Create an HTML page.
- Write JavaScript code inside script tags.
- Execute the webpage in a browser.
- Verify the results.

5. Implementation with Code

Question 1: Display Message

- Create a webpage that uses JavaScript to display: Welcome to Emerging Technologies Information Portal



Code:

```
<!DOCTYPE html>

<html>

<head>

  <title>JavaScript Output Demo</title>

</head>

<body>

  <h1>JavaScript</h1>

  <h2>Welcome to JavaScript</h2>
```

```
<script>

    alert("Welcome to Emerging Technologies Information Portal");

</script>
```

```
</body>
```

```
</html>
```

Question 2: Perform Simple Mathematical Operations

Create a webpage that accepts two numbers and performs:

- Addition
- Subtraction
- Multiplication
- Division

Mathematical Operations Using JavaScript

Enter First Number:

Enter Second Number:

Addition = 28

Subtraction = 22

Multiplication = 75

Division = 8.333333333333334

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Mathematical Operations</title>
```

```
</head>
```

```
<body>
```

<h1>Mathematical Operations Using JavaScript</h1>

<label>Enter First Number:</label>

<input type="number" id="num1">

<label>Enter Second Number:</label>

<input type="number" id="num2">

<button onclick="calculate()">Calculate</button>

<h2 id="add"></h2>

<h2 id="sub"></h2>

<h2 id="mul"></h2>

<h2 id="div"></h2>

<script>

function calculate()

{

var a = Number(document.getElementById("num1").value);

var b = Number(document.getElementById("num2").value);

document.getElementById("add").innerHTML = "Addition = " + (a + b);

document.getElementById("sub").innerHTML = "Subtraction = " + (a - b);

document.getElementById("mul").innerHTML = "Multiplication = " + (a * b);

document.getElementById("div").innerHTML = "Division = " + (a / b);

}

</script>

</body>

</html>

Question 3: Display Current Date and Time

- Write a JavaScript program to display the current date and time on the webpage.

Current Date and Time

Display Date and Time

**Current Date and Time: Fri Jun 19 2026 11:18:51
GMT+0530 (India Standard Time)**

Code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Current Date and Time</title>

</head>

<body>

  <h1>Current Date and Time</h1>

  <button onclick="showDateTime()">Display Date and Time</button>

  <h2 id="result"></h2>

  <script>

    function showDateTime()

    {

      var today = new Date();

      document.getElementById("result").innerHTML =
```

```

        "Current Date and Time: " + today;
    }
</script>

</body>

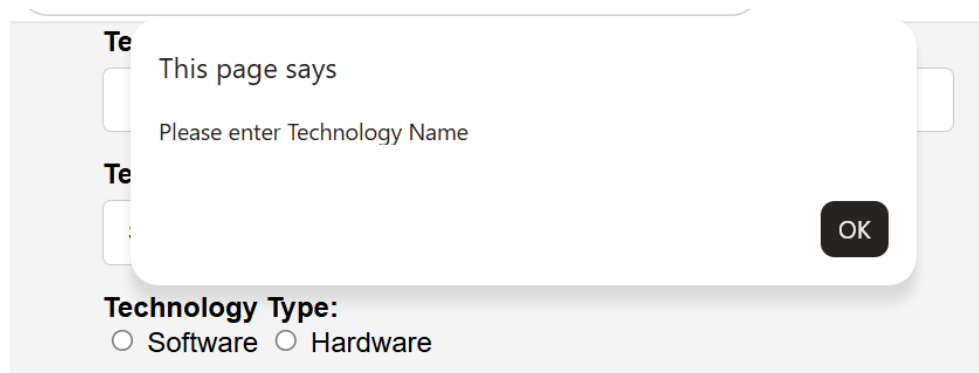
</html>

```

Question 4: Validate Technology Name

- Modify the Technology Suggestion Form and validate that:
 - Technology Name cannot be empty.

Display an alert message if the field is left blank.



Code:

```

<!DOCTYPE html>

<html>

<head>

    <title>Technology Suggestion Form</title>

</head>

<script>

    function validateForm()

    {

```

```
var tech = document.getElementById("techname").value;
```

```
if (tech == "")
```

```
{
```

```
    alert("Please enter Technology Name");
```

```
    return false;
```

```
}
```

```
    alert("Form Submitted Successfully");
```

```
    return true;
```

```
}
```

```
</script>
```

```
</head>
```

```
<body>
```

```
<h2>Technology Suggestion Form</h2>
```

```
<form onsubmit="return validateForm()">
```

```
<label>Technology Name:</label><br>
```

```
<input type="text" id="techname"><br><br>
```

```
<label>Technology Description:</label><br>
```

```
<textarea rows="4" cols="30"></textarea><br><br>
```

```
<label>Technology Type:</label><br>
```

```
<input type="radio" name="type"> Software
```

```
<input type="radio" name="type"> Hardware
```

```
<br><br>
```

```
<input type="submit" value="Submit">
```

```
</form>
```

```
</body>
```

```
</html>
```

Question 5: Validate Email Address

Add JavaScript validation to ensure:

- Email field is not empty.
- Email contains '@'.

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Email Validation</title>
```

```
<script>
```

```
function validateEmail()
```

```
{
```

```
    var email = document.getElementById("email").value;
```

```
    if(email == "")
```

```
    {
```

```
        alert("Please enter Email Address");
```

```
        return false;
```

```
    }
```

```
        if(email.indexOf("@") == -1)
        {
            alert("Please enter a valid Email Address");

            return false;
        }

        alert("Email is Valid");

        return true;
    }
</script>
</head>

<body>

    <h2>Email Validation Form</h2>

    <form onsubmit="return validateEmail()">

        <label>Email:</label><br>

        <input type="text" id="email"><br><br>

        <input type="submit" value="Submit">

    </form>

</body>
</html>
```

Question 6: Validate Contact Number

Add validation to ensure:

- Contact Number is not empty.

- Contact Number contains exactly 10 digits.

Code:

```
<!DOCTYPE html>

<html>

<head>

  <title>Contact Number Validation</title>


<script>

  function validateContact()

  {

    var contact = document.getElementById("contact").value;


    if(contact == "")

    {

      alert("Please enter Contact Number");

      return false;

    }


    if(contact.length != 10)

    {

      alert("Contact Number must contain exactly 10 digits");

      return false;

    }


    alert("Contact Number is Valid");

    return true;

  }

</script>
```

```
</head>
```

```
<body>
```

```
<h2>Contact Number Validation Form</h2>
```

```
<form onsubmit="return validateContact()">
```

```
<label>Contact Number:</label><br>
```

```
<input type="text" id="contact"><br><br>
```

```
<input type="submit" value="Submit">
```

```
</form>
```

```
</body>
```

```
</html>
```

Question 7: Validate Technology Category

- Ensure that the user selects a category from the dropdown list.

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Technology Category Validation</title>
```

```
<script>
```

```
function validateCategory()
{
    var category = document.getElementById("category").value;

    if(category == "")
    {
        alert("Please select Technology Category");
        return false;
    }

    alert("Technology Category Selected");
    return true;
}
</script>
</head>
```

```
<body>
```

```
<h2>Technology Category Form</h2>
```

```
<form onsubmit="return validateCategory()">
```

```
<label>Technology Category:</label><br><br>
```

```
<select id="category">
```

```
<option value="">-- Select Category --</option>
```

```
<option>Artificial Intelligence</option>
```

```
<option>Cloud Computing</option>
```

```
<option>Cyber Security</option>
```

```
<option>Internet of Things</option>
```

```
</select>
```

```
<br><br>
```

```
<input type="submit" value="Submit">
```

```
</form>
```

```
</body>
```

```
</html>
```

Question 8: Validate Radio Buttons

Ensure that the user selects a Technology Type:

- Software
- Hardware

Display an appropriate error message if no option is selected.

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Radio Button Validation</title>
```

```
<script>
```

```
function validateType()
```

```
{
```

```
    var software = document.getElementById("software").checked;
```

```
    var hardware = document.getElementById("hardware").checked;
```

```
    if(!software && !hardware)
```

```
{
```

```
    alert("Please select Technology Type");  
    return false;  
}
```

```
    alert("Technology Type Selected");  
    return true;  
}
```

```
</script>
```

```
</head>
```

```
<body>
```

```
<h2>Technology Type Form</h2>
```

```
<form onsubmit="return validateType()">
```

```
<label>Technology Type:</label><br><br>
```

```
<input type="radio" id="software" name="type">
```

```
<label for="software">Software</label>
```

```
<input type="radio" id="hardware" name="type">
```

```
<label for="hardware">Hardware</label>
```

```
<br><br>
```

```
<input type="submit" value="Submit">
```

```
</form>
```

</body>

</html>

Question 9: Validate Checkboxes

Ensure that the user selects at least one Application Area.

Options:

- Education
- Healthcare
- Finance
- Industry

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Checkbox Validation</title>
```

```
  <script>
```

```
    function validateApplication()
```

```
    {
```

```
      var education = document.getElementById("education").checked;
```

```
      var healthcare = document.getElementById("healthcare").checked;
```

```
      var finance = document.getElementById("finance").checked;
```

```
      var industry = document.getElementById("industry").checked;
```

```
      if(!education && !healthcare && !finance && !industry)
```

```
      {
```

```
        alert("Please select at least one Application Area");
```

```
        return false;
    }

    alert("Application Area Selected");

    return true;
}
</script>
</head>

<body>

    <h2>Application Area Form</h2>

    <form onsubmit="return validateApplication()">

        <label>Application Area:</label><br><br>

        <input type="checkbox" id="education">
        <label for="education">Education</label><br>

        <input type="checkbox" id="healthcare">
        <label for="healthcare">Healthcare</label><br>

        <input type="checkbox" id="finance">
        <label for="finance">Finance</label><br>

        <input type="checkbox" id="industry">
        <label for="industry">Industry</label><br><br>

        <input type="submit" value="Submit">
```

</form>

</body>

</html>

6. Result

Successfully performed all the questions.

7. Applications

- Client-side validation
- Interactive web applications
- Dynamic webpage development

8. Practice Questions

1. Student Result Calculator

Accept marks of three subjects from the user and calculate:

- Total Marks
- Percentage
- Display the result on the webpage.

2. Employee Information Form

Create a form containing:

- Employee Name
- Email
- Contact Number
- Department

Validate all fields using JavaScript.

9. Questions (Viva / Oral)

1. What is JavaScript?
2. Difference between Java and JavaScript?
3. What is a variable?
4. What is a function?
5. What is alert()?
6. What is form validation?
7. Difference between == and ===?
8. What is an event?
9. What is the DOM?
10. What are the advantages of JavaScript?

PRACTICAL 6

1. Aim

PHP scripts to perform simple mathematical operations and Retrieving data from HTML forms

2. Prior Concepts

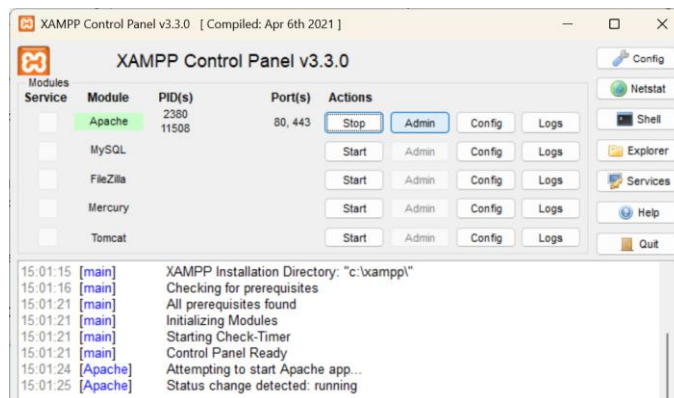
- HTML Forms
- Basic JavaScript
- Web Browser and Text Editor

3. New Concepts

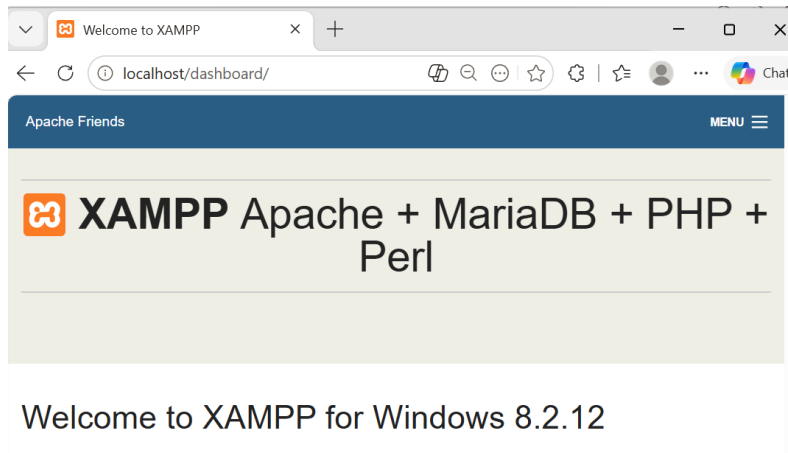
- PHP Syntax
- Variables and Operators
- Sending Request to Server
- Receiving Response from Server
- GET and POST Methods

4. Procedure

- Create folder on the server under - C:/XAMP/htdocs/www.
- Store all PHP scripts and HTML pages in the same folder.
- Start XAMP Server.



- open browser and type – "localhost"



- Open the file to be executed through server
- Observe the output.

5. Implementation with Code

Question 1: Display a Message Using PHP

Create a PHP script to display a welcome message for the Emerging Technologies Information Portal.

Code:

```
<?php  
  
echo "<h2>Welcome to the Emerging Technologies Information Portal</h2>";  
  
echo "<p>Explore the latest trends in Artificial Intelligence, Cloud Computing, IoT, Blockchain, and Cyber Security.</p>";  
  
?>
```

Question 2: Perform Simple Mathematical Operations

Write a PHP program to perform Addition, Subtraction, Multiplication and Division of two numbers.

Code:

```
<?php  
  
$num1 = 20;  
$num2 = 10;  
  
echo "<h2>Mathematical Operations</h2>";  
  
echo "First Number = $num1 <br>";  
echo "Second Number = $num2 <br><br>";
```

```
echo "Addition = " . ($num1 + $num2) . "<br>";  
echo "Subtraction = " . ($num1 - $num2) . "<br>";  
echo "Multiplication = " . ($num1 * $num2) . "<br>";  
echo "Division = " . ($num1 / $num2) . "<br>";  
?>
```

Question 3: Create a Simple Calculator Using PHP

Create a webpage that accepts two numbers(using HTML Form) and displays mathematical operations using PHP.

Code:

```
<!DOCTYPE html>  
  
<html>  
  
<head>  
    <title>Simple PHP Calculator</title>  
</head>  
  
<body>  
  
    <h2>Simple Calculator</h2>  
  
    <form method="post">  
        Enter First Number:  
        <input type="number" name="num1" required><br><br>  
  
        Enter Second Number:  
        <input type="number" name="num2" required><br><br>  
  
        <input type="submit" value="Calculate">  
    </form>  
  
    <?php  
    if ($_SERVER["REQUEST_METHOD"] == "POST") {
```

```
$num1 = $_POST['num1'];  
$num2 = $_POST['num2'];  
  
echo "<h3>Result</h3>";  
  
echo "Addition = " . ($num1 + $num2) . "<br>";  
echo "Subtraction = " . ($num1 - $num2) . "<br>";  
echo "Multiplication = " . ($num1 * $num2) . "<br>";  
  
if ($num2 != 0) {  
    echo "Division = " . ($num1 / $num2) . "<br>";  
} else {  
    echo "Division = Cannot divide by zero.";  
}  
}  
?>  
</body>  
</html>
```

Question 4: Process Technology Suggestion Form Using PHP

Create a PHP page to receive and display information submitted through the Technology Suggestion Form.

Code:

```
<!DOCTYPE html>  
<html>  
<head>  
    <title>Technology Suggestion Form</title>  
</head>  
<body>  
  
<h2>Technology Suggestion Form</h2>
```

```
<form method="post">
```

Name:

```
<input type="text" name="name" required><br><br>
```

Email:

```
<input type="email" name="email" required><br><br>
```

Favorite Technology:

```
<input type="text" name="technology" required><br><br>
```

Suggestion:

```
<textarea name="suggestion" rows="4" cols="30" required></textarea><br><br>
```

```
<input type="submit" value="Submit">
```

```
</form>
```

```
<?php
```

```
if ($_SERVER["REQUEST_METHOD"] == "POST") {
```

```
    $name = $_POST['name'];
```

```
    $email = $_POST['email'];
```

```
    $technology = $_POST['technology'];
```

```
    $suggestion = $_POST['suggestion'];
```

```
    echo "<h2>Submitted Information</h2>";
```

```
    echo "Name: $name <br>";
```

```
echo "Email: $email <br>";  
  
echo "Favorite Technology: $technology <br>";  
  
echo "Suggestion: $suggestion <br>";  
  
}  
  
?>  
  
</body>  
  
</html>
```

6. Result

Successfully performed all the questions.

7. Applications

- Dynamic Web Pages
- Server-side Calculations
- Business Applications

8. Practice Questions

- Write a PHP script to display your: Name, Class, College Name
- Create a form to accept two numbers and display the largest number using PHP.
- Create a Student Information Form using HTML and PHP. The form should collect the following information from the student and display the submitted details on another PHP page.

Fields to Include

1. Student Name (Text Box)
2. Roll Number (Text Box)
3. Email Address (Email Control)
4. Contact Number (Text Box / Tel Control)
5. Gender (Radio Buttons)
 - Male
 - Female
 - Other
6. Course (Dropdown List)
 - B.Sc. Information Technology
 - B.Sc. Computer Science
 - B.Sc. Data Science
7. Skills (Checkboxes)
 - HTML
 - CSS
 - JavaScript
 - PHP
 - MySQL
8. Date of Birth (Date Control)
9. Address (Text Area)

10. Submit Button

11. Reset Button

9. Viva Questions

1. What is PHP?
2. What does PHP stand for?
3. Why is PHP called server-side scripting language?
4. What is XAMPP?
5. What is localhost?
6. How do you declare variables in PHP?
7. What is echo statement?
8. What is GET method?
9. What is POST method?
10. Difference between GET and POST?

PRACTICAL 7

1. Aim

PHP to work with Arrays and Files (Reading / Writing).

2. Prior Concepts

- HTML Forms
- PHP Variables
- Form Controls
- GET and POST Methods

3. New Concepts

- Indexed Arrays
- Associative Arrays
- File Handling Functions
- Reading and Writing Files

4. Procedure

- Create HTML form.
- Create PHP file to process form data.
- Connect form with PHP script.
- Execute and verify output.

5. Implementation with Code

Question 1: Indexed Array in PHP

Create a PHP program to store the technology categories in an indexed array and display all categories using a foreach loop. Also display the total number of categories using the count() function.

Code:

```
<?php
```

```
// Indexed Array
```

```
$technologies = array(
```

```
    "Artificial Intelligence",
```

```
    "Cloud Computing",
```

```
    "Internet of Things",
```

```
    "Blockchain",
```

```
    "Cyber Security"
```

```
);
```

```
echo "<h2>Technology Categories</h2>";
```

```
// Display using foreach loop
```

```
foreach ($technologies as $technology) {  
    echo $technology . "<br>";  
}
```

```
// Display total number of categories
```

```
echo "<br>Total Categories: " . count($technologies);  
?>
```

Question 2: Associative Array in PHP

Create an associative array to store technology information and display the values.

Store the following details and display all values using a foreach loop.

Technology Information

Field	Value
Technology Name	Artificial Intelligence
Category	Software Technology
Year Introduced	2023
Application Area	Healthcare
Developer	OpenAI
Programming Language	Python
Website	www.openai.com

Code:

```
<?php
```

```
// Associative Array
```

```
$technology = array(  
    "Technology Name" => "Artificial Intelligence",  
    "Category" => "Software Technology",  
    "Year Introduced" => "2023",
```

```
"Application Area" => "Healthcare",  
"Developer" => "OpenAI",  
"Programming Language" => "Python",  
"Website" => "www.openai.com"  
);
```

```
echo "<h2>Technology Information</h2>";
```

```
// Display using foreach loop  
foreach ($technology as $key => $value) {  
    echo "<b>$key:</b> $value <br>";  
}  
?>
```

Question 3: Write to a file

Write a PHP program to save the submitted Technology Suggestion Form data into a text file(technology.txt).

suggest.html → JavaScript Validation → save.php → Data Saved in technology.txt

Code:

Suggest.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Technology Suggestion Form</title>
```

```
<script>
```

```
function validateForm() {  
    let name = document.forms["techForm"]["name"].value;  
    let email = document.forms["techForm"]["email"].value;  
    let tech = document.forms["techForm"]["technology"].value;  
    let suggestion = document.forms["techForm"]["suggestion"].value;
```

```
if (name == "" || email == "" || tech == "" || suggestion == "") {  
    alert("All fields are required!");  
    return false;  
}
```

```
return true;  
}
```

```
</script>
```

```
</head>
```

```
<body>
```

```
<h2>Technology Suggestion Form</h2>
```

```
<form name="techForm" action="save.php" method="post" onsubmit="return validateForm()">
```

Name:

```
<input type="text" name="name"><br><br>
```

Email:

```
<input type="email" name="email"><br><br>
```

Technology:

```
<input type="text" name="technology"><br><br>
```

Suggestion:

```
<textarea name="suggestion"></textarea><br><br>
```

```
<input type="submit" value="Submit">
```

</form>

</body>

</html>

Save.php

<?php

```
$name = $_POST['name'];
```

```
$email = $_POST['email'];
```

```
$technology = $_POST['technology'];
```

```
$suggestion = $_POST['suggestion'];
```

```
$data = "Name: $name\n";
```

```
$data .= "Email: $email\n";
```

```
$data .= "Technology: $technology\n";
```

```
$data .= "Suggestion: $suggestion\n";
```

```
$data .= "-----\n";
```

```
$file = fopen("technology.txt", "a");
```

```
fwrite($file, $data);
```

```
fclose($file);
```

```
echo "<h2>Data Saved Successfully!</h2>";
```

```
?>
```

Question 4: Read from a file

Read and display the contents of technology.txt.

suggest.html → JavaScript Validation → save.php → Data Saved in technology.txt → read.php → display data on the web page

Code:

Read.php

```
<?php
```

```
echo "<h2>Technology Suggestions</h2>";
```

```
$file = fopen("technology.txt", "r");
```

```
while (!feof($file)) {  
    echo fgets($file) . "<br>";  
}
```

```
fclose($file);
```

```
?>
```

6. Result

Successfully performed all the questions.

7. Applications

- Data Storage
- File Processing
- Report Generation

8. Practice Questions

1. Create an indexed array containing:

- HTML
- CSS
- JavaScript
- PHP

- MySQL

Display all elements using foreach.

2. Create an associative array storing:

- Student Name
- Roll Number
- Course
- Email

Display all values.

3. Design and develop a **Technology Portal Feedback Form** using HTML and PHP. The form should allow visitors to provide feedback about the **Emerging Technologies Information Portal**.

Form Fields

The form should contain the following controls:

- i. Visitor Name (Text Box)
- ii. Email Address (Email Control)
- iii. Technology Category Visited (Dropdown List)
 - Artificial Intelligence
 - Cloud Computing
 - Cyber Security
 - Internet of Things (IoT)
 - Blockchain
- iv. Overall Rating (Radio Buttons)
 - Excellent
 - Good
 - Average
 - Poor
- v. Features Liked (Checkboxes)
 - Design
 - Content
 - Navigation
 - Technology Information
- vi. Suggestions / Comments (Text Area)
- vii. Submit Button

viii. Reset Button

- Submit the form data to a PHP page using the POST method.
- Retrieve all submitted values using `$_POST`.
- Open a text file named `feedback.txt` in append mode.
- Store the submitted feedback in the text file in a readable format.
- Display a message:
- "Feedback Saved Successfully"

9. Viva Questions

1. What is an array?
2. Types of arrays in PHP?
3. What is an associative array?
4. What is a multidimensional array?
5. What is `fopen()`?
6. What is `fclose()`?
7. What is `fread()`?
8. What is `fwrite()`?
9. What is file handling?
10. Advantages of arrays?

PRACTICAL 8

1. Aim

Advanced PHP: Sessions, Cookies, Filters

2. Prior Concepts

- PHP Variables
- Forms
- Arrays

3. New Concepts

- Sessions
- Cookies
- Filters
- Session Management

4. Procedure

- Create PHP scripts.
- Implement sessions and cookies.
- Apply filters.
- Execute and verify output.

5. Implementation with Code

Question 1: Sessions

Create a Login Form for the Emerging Technologies Information Portal. After successful login, store the username in a session and display the Technology Suggestion Form. Unauthorized users should not be able to access the form directly.

login.html → login.php → Session Created → suggest.html → logout.php

1. login.html

User enters:

- Username
- Password

2. login.php

- Validates credentials
- Creates session

3. suggest.php

- Checks session
- Displays Technology Suggestion Form

4. logout.php

- Destroys session
- Returns to Login Page

1. login.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Login Page</title>
```

```
</head>
```

```
<body>
```

```
<h2>Emerging Technologies Information Portal</h2>
```

```
<form action="login.php" method="post">
```

Username:

```
<input type="text" name="username" required><br><br>
```

Password:

```
<input type="password" name="password" required><br><br>
```

```
<input type="submit" value="Login">
```

```
</form>
```

```
</body>
```

```
</html>
```

2. login.php

```
<?php
```

```
session_start();
```

```
$username = $_POST['username'];  
$password = $_POST['password'];  
  
// Valid Username and Password  
if ($username == "admin" && $password == "12345")  
{  
    $_SESSION['username'] = $username;  
  
    header("Location: suggest.php");  
}  
else  
{  
    echo "<h2>Invalid Username or Password!</h2>";  
    echo "<a href='login.html'>Try Again</a>";  
}  
?>
```

3. suggest.php

```
<?php  
session_start();  
  
// Check Session  
if (!isset($_SESSION['username']))  
{  
    header("Location: login.html");  
    exit();  
}  
?>
```

```
<!DOCTYPE html>
```

```
<html>

<head>

    <title>Technology Suggestion Form</title>

</head>

<body>


<h2>Welcome,

<?php
echo $_SESSION['username'];
?>

</h2>


<h3>Technology Suggestion Form</h3>


<form>


Name:

<input type="text"><br><br>


Email:

<input type="email"><br><br>


Technology:

<input type="text"><br><br>


Suggestion:

<textarea rows="5" cols="30"></textarea><br><br>


<input type="submit" value="Submit">
```

```
</form>
```

```
<br>
```

```
<a href="logout.php">Logout</a>
```

```
</body>
```

```
</html>
```

4. logout.php

```
<?php
```

```
session_start();
```

```
// Destroy Session
```

```
session_destroy();
```

```
// Redirect to Login Page
```

```
header("Location: login.html");
```

```
exit();
```

```
?>
```

Question 2: Cookies

- **Store the user's preferred technology category in a cookie**
- **Display it when the user revisits the portal**

1. category.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Select Technology Category</title>
```

```
</head>
```

```
<body>
```

```
<h2>Select Your Preferred Technology Category</h2>
```

```
<form action="setcookie.php" method="post">
```

Preferred Technology:

```
<select name="category">
```

```
<option>Artificial Intelligence</option>
```

```
<option>Cloud Computing</option>
```

```
<option>Cyber Security</option>
```

```
<option>Blockchain</option>
```

```
<option>Internet of Things</option>
```

```
</select>
```

```
<br><br>
```

```
<input type="submit" value="Save Preference">
```

```
</form>
```

```
</body>
```

```
</html>
```

2. setcookie.php

```
<?php
```

```
$category = $_POST['category'];
```

```
// Create Cookie (Valid for 1 day)
```

```
setcookie("technology", $category, time() + 86400);
```

```
echo "<h2>Your Preference has been Saved Successfully!</h2>";
```

```
echo "<a href='welcome.php'>Visit Portal</a>";
```

```
?>
```

3. welcome.php

```
<?php
```

```
?>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Emerging Technologies Information Portal</title>
```

```
</head>
```

```
<body>
```

```
<h2>Welcome to Emerging Technologies Information Portal</h2>
```

```
<?php
```

```
if(isset($_COOKIE['technology']))
```

```
{
```

```
    echo "<h3>Your Preferred Technology Category is: "
```

```
        . $_COOKIE['technology'] . "</h3>";
```

```
}
```

```
else
```

```
{
```

```
    echo "<h3>No Preference Found.</h3>";
```

```
}
```

?>

</body>

</html>

Question 3: Validate and Sanitize User Input Using PHP Filters

Create a PHP program to accept user information through a form and perform validation and sanitization using appropriate PHP Filters.

Form Fields

1. Name
2. Email Address
3. Age
4. Website URL
5. IP Address

Filter	Purpose
<code>FILTER_SANITIZE_STRING</code>	Remove HTML tags and sanitize user input
<code>FILTER_VALIDATE_EMAIL</code>	Validate Email Address
<code>FILTER_VALIDATE_INT</code>	Validate Integer Values
<code>FILTER_VALIDATE_URL</code>	Validate Website URL
<code>FILTER_VALIDATE_IP</code>	Validate IP Address

Code:

1. index.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>PHP Filters Validation</title>
```

```
</head>
```

```
<body>
```

```
<h2>User Information Form</h2>
```

```
<form action="validate.php" method="post">
```

Name:

```
<input type="text" name="name" required>
```

```
<br><br>
```

Email Address:

```
<input type="email" name="email" required>
```

```
<br><br>
```

Age:

```
<input type="text" name="age" required>
```

```
<br><br>
```

Website URL:

```
<input type="text" name="website" required>
```

```
<br><br>
```

IP Address:

```
<input type="text" name="ip" required>
```

```
<br><br>
```

```
<input type="submit" value="Submit">
```

```
</form>
```

```
</body>
```

```
</html>
```

2. validate.php

```
<?php
```

```
// Sanitize Name
```

```
$name = filter_var($_POST['name'], FILTER_SANITIZE_FULL_SPECIAL_CHARS);
```

```
// Get Form Data
```

```
$email = $_POST['email'];
```

```
$age = $_POST['age'];
```

```
$website = $_POST['website'];
```

```
$ip = $_POST['ip'];
```

```
echo "<h2>Validation Result</h2>";
```

```
// Display Sanitized Name
```

```
echo "<b>Name:</b> $name <br><br>";
```

```
// Validate Email
```

```
if(filter_var($email, FILTER_VALIDATE_EMAIL))
```

```
{
```

```
    echo "Email: Valid<br>";
```

```
}
```

```
else
```

```
{
```

```
    echo "Email: Invalid<br>";
```

```
}
```

```
// Validate Age
```

```
if(filter_var($age, FILTER_VALIDATE_INT))
```

```
{
```

```
    echo "Age: Valid<br>";
```

```
}  
  
else  
  
{  
  
    echo "Age: Invalid<br>";  
  
}  
  
  
// Validate Website  
  
if(filter_var($website, FILTER_VALIDATE_URL))  
  
{  
  
    echo "Website: Valid<br>";  
  
}  
  
else  
  
{  
  
    echo "Website: Invalid<br>";  
  
}  
  
  
// Validate IP Address  
  
if(filter_var($ip, FILTER_VALIDATE_IP))  
  
{  
  
    echo "IP Address: Valid<br>";  
  
}  
  
else  
  
{  
  
    echo "IP Address: Invalid<br>";  
  
}  
  
  
?>
```

6. Result

Successfully performed all the questions.

7. Applications

- User Authentication
- User Tracking
- Security

8. Practice Questions

1. Create a PHP program to create a cookie named technology and then provide an option to delete the cookie.
2. Create multiple cookies and display all stored cookies using the `$_COOKIE` array.
3. Allow the user to select a theme (Light/Dark). Store the selected theme in a cookie and display the selected theme when the page is revisited.

9. Viva Questions

1. What is a session?
2. What is a cookie?
3. Difference between session and cookie?
4. What is `session_start()`?
5. What is `setcookie()`?
6. What is `filter_var()`?
7. Why are sessions used?
8. Where are cookies stored?
9. How can cookies be deleted?
10. Advantages of filters?

PRACTICAL 9

1. Aim

PHP and MySQL: Create a Database and Tables

2. Prior Concepts

- PHP Basics
- Database Concepts

3. New Concepts

- MySQL Database
- SQL Commands
- Database Connectivity
- Table Creation

4. Procedure

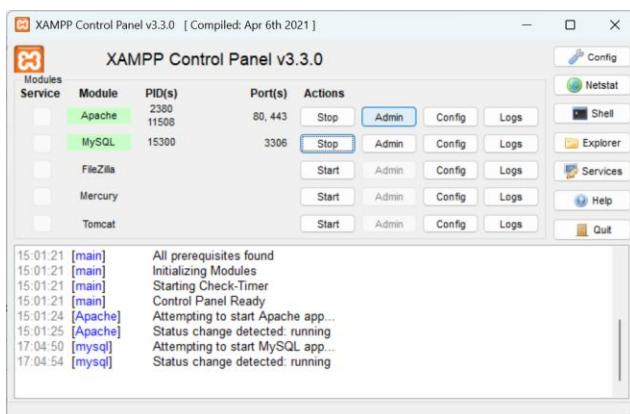
- Start Apache and MySQL services in XAMPP Control Panel.
- Open a web browser.
- Enter `http://localhost/phpmyadmin`.

5. Implementation with Code

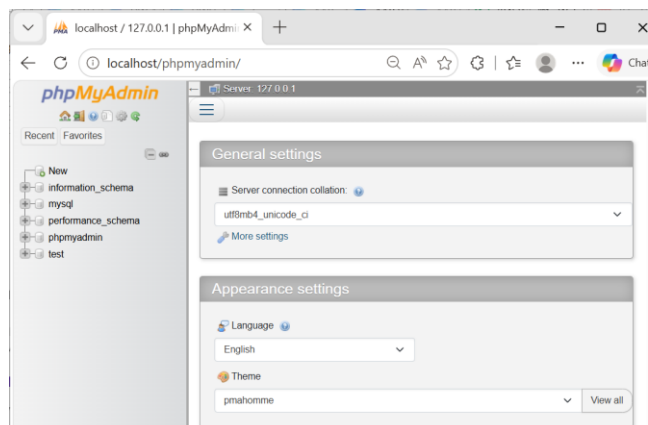
Question 1: Create a MySQL database named TechnologyPortal for the Emerging Technologies Information Portal.

- Start Apache and MySQL services in XAMPP Control Panel.
- Open a web browser.
- Enter `http://localhost/phpmyadmin`.
- Click New from the left panel.
- Enter database name: TechnologyPortal.
- Click Create.
- Verify that the database appears in the left navigation panel.

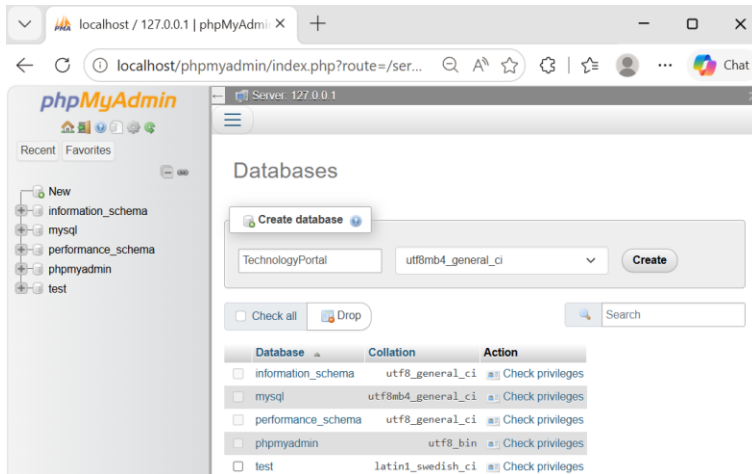
APACHE Server and MySQL



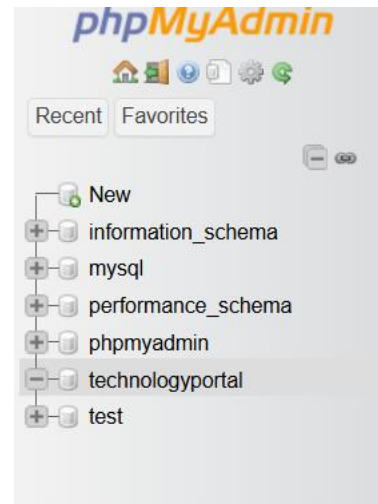
phpMyAdmin



Create Database



technologyportal database



Code:

Steps:

1. Open the **XAMPP Control Panel**.
2. Start the **Apache** and **MySQL** services by clicking the **Start** button beside each service.
3. Open a web browser such as Google Chrome or Microsoft Edge.
4. Enter the following URL in the address bar:

<http://localhost/phpmyadmin>

5. Press **Enter** to open phpMyAdmin.
6. Click **New** in the left navigation panel.
7. In the **Database name** field, enter:

TechnologyPortal

8. Keep the default character set and collation, or select **utf8mb4_general_ci**.
9. Click **Create**.
10. Verify that the **TechnologyPortal** database appears in the left navigation panel

Question 2: Create a Table

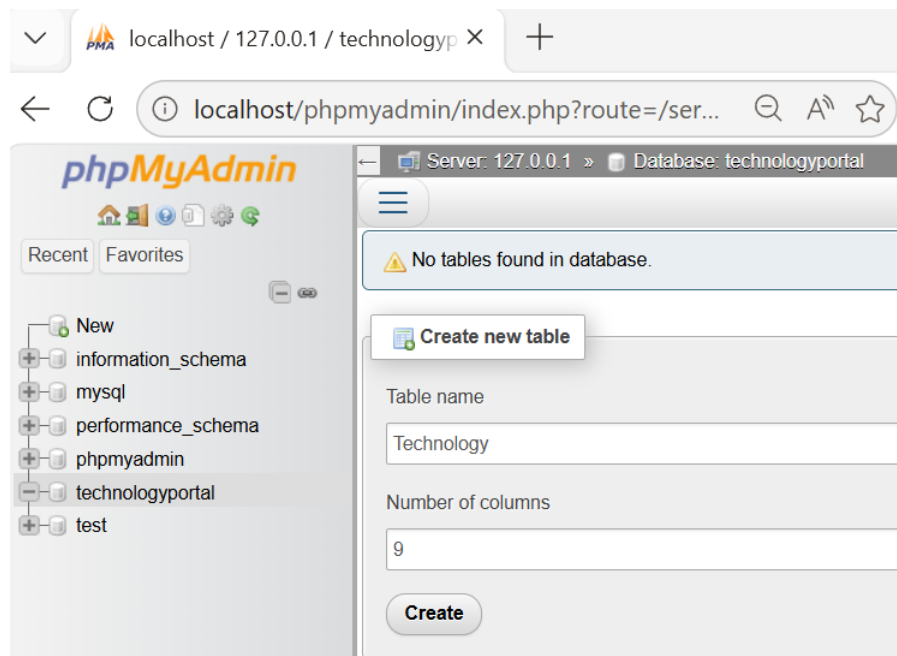
Create a table named **Technology** in the **TechnologyPortal** database to store technology information submitted through the **Technology Suggestion Form**.

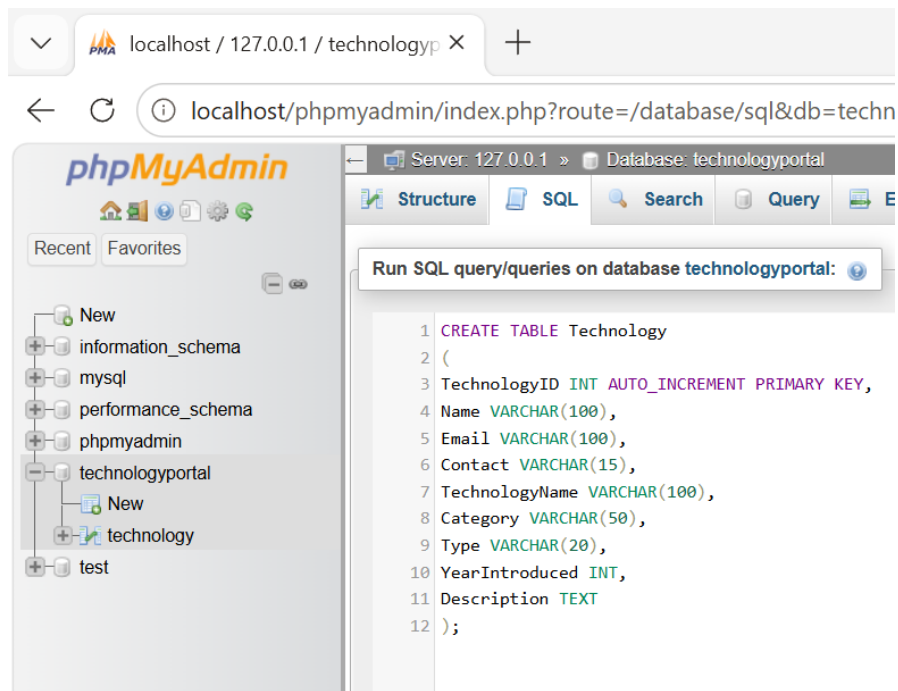
Table Structure:

TechnologyID INT AUTO_INCREMENT PRIMARY KEY
Name VARCHAR(100)
Email VARCHAR(100)
Contact VARCHAR(15)
TechnologyName VARCHAR(100)
Category VARCHAR(50)
Type VARCHAR(20)
YearIntroduced INT
Description TEXT

Steps:

1. Select TechnologyPortal database.
2. Click SQL tab.
3. Enter the CREATE TABLE statement.
4. Click Go.
5. Verify that the Technology table appears under the database.





<?php

// Database Connection

\$servername = "localhost";

\$username = "root";

\$password = "";

\$dbname = "TechnologyPortal";

// Create Connection

\$conn = mysqli_connect(\$servername, \$username, \$password, \$dbname);

// Check Connection

if (!\$conn)

{

die("Connection Failed: " . mysqli_connect_error());

}

// Get Form Data

\$name = \$_POST['name'];

```

$email = $_POST['email'];

$contact = $_POST['contact'];

$technologyName = $_POST['technologyName'];

$category = $_POST['category'];

$type = $_POST['type'];

$yearIntroduced = $_POST['yearIntroduced'];

$description = $_POST['description'];


// SQL Query

$sql = "INSERT INTO Technology
(Name, Email, Contact, TechnologyName, Category, Type, YearIntroduced, Description)

VALUES
('$name', '$email', '$contact', '$technologyName', '$category', '$type', '$yearIntroduced', '$description')";


// Execute Query

if(mysqli_query($conn, $sql))
{
    echo "<h2>Record Inserted Successfully!</h2>";
}
else
{
    echo "Error: " . mysqli_error($conn);
}


// Close Connection

mysqli_close($conn);

?>

```

Question 3: Display Database and Table Structure

Display the list of databases, tables, and structure of the Technology table.

- Select TechnologyPortal database.
- Click SQL tab.
- Execute the commands to display all databases, tables and table description.
- Observe the output.

Steps

1. Open **phpMyAdmin**.
2. Select the **TechnologyPortal** database from the left navigation panel.
3. Click the **SQL** tab.
4. Execute the following SQL commands one by one.
5. Observe the output displayed in phpMyAdmin.

SQL Commands

1. Display all Databases

```
SHOW DATABASES;
```

2. Display all Tables in the TechnologyPortal Database

```
USE TechnologyPortal;
```

```
SHOW TABLES;
```

3. Display the Structure of the Technology Table

```
DESCRIBE Technology;
```

or

```
DESC Technology;
```

Expected Output

Output of SHOW DATABASES

A list of all databases available in the MySQL server, including:

- information_schema
- mysql
- performance_schema
- phpmyadmin
- **TechnologyPortal**

Output of SHOW TABLES

The **TechnologyPortal** database should contain the following table:

Tables_in_TechnologyPortal

Technology

Output of DESCRIBE Technology

Question 4: Insert Sample Records

Insert at least three technology records into the **Technology** table.

Steps:

1. Select Technology table.
2. Click Insert tab.
3. Enter values for all fields except TechnologyID.
4. Click Go.
5. Repeat for additional records.

USE TechnologyPortal;

INSERT INTO Technology

(Name, Email, Contact, TechnologyName, Category, Type, YearIntroduced, Description)

VALUES

('Rahul Sharma', 'rahul@gmail.com', '9876543210',

'Artificial Intelligence', 'Artificial Intelligence',

'Software', 1956,

'Technology that enables machines to simulate human intelligence.'),

('Priya Patel', 'priya@gmail.com', '9123456780',

'Blockchain', 'Blockchain',

'Platform', 2008,

'Distributed ledger technology used for secure and transparent transactions.'),

```
('Amit Verma', 'amit@gmail.com', '9988776655',  
'Internet of Things', 'Internet of Things',  
'Hardware', 1999,  
'Network of physical devices connected to the Internet for data exchange.');
```

Verify the Inserted Records

Execute the following SQL query:

```
SELECT * FROM Technology;
```

Expected Output

Question 5: Display Records

Retrieve and display all records stored in the Technology table.

Steps:

1. Select Technology table.
2. Click SQL tab.
3. Execute the SELECT query.
4. Observe the displayed records.

Steps

1. Open **phpMyAdmin**.
2. Select the **TechnologyPortal** database.
3. Select the **Technology** table from the left navigation panel.
4. Click the **SQL** tab.
5. Enter the following SQL query.
6. Click **Go** to execute the query.
7. Observe the displayed records.

SQL Query

```
USE TechnologyPortal;
```

```
SELECT * FROM Technology;
```

6. Result

Successfully performed all the questions.

7. Applications

- Database Management
- Data Storage
- Web Applications

8. Practice Questions

1. A Training Institute wants to store the following information:

- Trainer Name
- Subject
- Experience
- Email
- Contact Number

Identify suitable data types and write the SQL statement to create the table.

2. Create a database named **CourseManagement** and a table named **Course** with the following fields:

| Field Name | Data Type |
|------------|--------------------------------|
| CourseID | INT AUTO_INCREMENT PRIMARY KEY |
| CourseName | VARCHAR(100) |
| Duration | VARCHAR(20) |
| Fees | INT |

Insert five courses and display the table structure.

3. Create a database named **StudentPortal** and a table named **Student** with the following fields:

| Field Name | Data Type |
|-------------|-----------------|
| RollNo | INT PRIMARY KEY |
| StudentName | VARCHAR(100) |
| Class | VARCHAR(20) |
| Email | VARCHAR(100) |
| Contact | VARCHAR(15) |

Insert three records and display all records.

9. Viva Questions

1. What is MySQL?
2. What is a database?
3. What is a table?

4. What is a primary key?
5. What is phpMyAdmin?
6. What is SQL?
7. What is CREATE DATABASE?
8. What is CREATE TABLE?
9. What is database connectivity?
10. Advantages of MySQL?

PRACTICAL 11

1. Aim

PHP and MySQL Integration: Perform CRUD Operations (Insert, Update, Delete, Display)

2. Prior Concepts

- PHP
- MySQL
- Database Connectivity

3. New Concepts

- CRUD Operations
- SQL Queries
- Dynamic Data Management

4. Procedure

- Create database connection.
- Create forms.
- Perform CRUD operations.
- Execute and verify output.

5. Implementation with Code

Question 1: Insert Technology Information

Create a PHP program to connect to the TechnologyPortal database and insert technology information submitted through the Technology Suggestion Form into the Technology table.

Steps:

1. Start Apache and MySQL from XAMPP Control Panel.
2. Open phpMyAdmin and ensure TechnologyPortal database exists.
3. Create insert.php file in htdocs.

4. Write PHP code to establish database connection.
5. Execute INSERT query.
6. Run the application using localhost.

1. Start **Apache** and **MySQL** services from the XAMPP Control Panel.
2. Open **phpMyAdmin** and ensure that the **TechnologyPortal** database and **Technology** table exist.
3. Create a file named **insert.php** inside the **htdocs/TechnologyPortal** folder.
4. Write the PHP code to establish a connection with the MySQL database.
5. Execute the **INSERT** query to store the submitted form data.
6. Run the application using the following URL:

...

<http://localhost/TechnologyPortal/technology.html>

...

PHP Code (insert.php)

```php

<?php

// Database Connection

\$servername = "localhost";

\$username = "root";

\$password = "";

\$dbname = "TechnologyPortal";

// Create Connection

\$conn = mysqli\_connect(\$servername, \$username, \$password, \$dbname);

// Check Connection

```
if (!$conn)

{

 die("Connection Failed: " . mysqli_connect_error());

}

// Receive Form Data

$name = $_POST["name"];

$email = $_POST["email"];

$contact = $_POST["contact"];

$technologyName = $_POST["technologyName"];

$category = $_POST["category"];

$type = $_POST["type"];

$yearIntroduced = $_POST["yearIntroduced"];

$description = $_POST["description"];

// SQL Query

$sql = "INSERT INTO Technology

(Name, Email, Contact, TechnologyName, Category, Type, YearIntroduced, Description)

VALUES

('$name', '$email', '$contact', '$technologyName', '$category', '$type', '$yearIntroduced', '$description')";

// Execute Query

if(mysqli_query($conn, $sql))

{

 echo "<h2>Technology information inserted successfully.</h2>";

}

else

{

 echo "Error: " . mysqli_error($conn);

}

// Close Connection

mysqli_close($conn);
```

?>

..

### ### HTML Form

Ensure the HTML form contains the following line:

```
```html
```

```
<form action="insert.php" method="post">
```

```
```
```

### ### Folder Structure

```
```
```

xampp

└─ htdocs

└─ TechnologyPortal

└─ technology.html

└─ insert.php

```
```
```

### ### Expected Output

\* The PHP program establishes a connection with the **TechnologyPortal** database.

\* When the user submits the **Technology Suggestion Form**, the entered details are inserted into the **Technology** table.

\* Upon successful insertion, the following message is displayed:

```
```
```

Technology information inserted successfully.

```
```
```

## Question 2: Display Technology Records

Create a PHP program to retrieve and display all technology records stored in the Technology table.

### Steps:

1. Create display.php.

2. Connect to MySQL database.
3. Execute SELECT \* query.
4. Fetch records using mysqli\_fetch\_assoc().
5. Display records in tabular format.

### PHP Code (display.php)

```
<?php

// Database Connection
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "TechnologyPortal";

// Create Connection
$conn = mysqli_connect($servername, $username, $password, $dbname);

// Check Connection
if (!$conn)
{
 die("Connection Failed: " . mysqli_connect_error());
}

// SQL Query
$sql = "SELECT * FROM Technology";

// Execute Query
$result = mysqli_query($conn, $sql);

?>

<!DOCTYPE html>
<html>
<head>
 <title>Technology Records</title>

 <style>
 body{
 font-family: Arial, sans-serif;
 }

 table{
 border-collapse: collapse;
 width:100%;
 }

 th,td{
 border:1px solid black;
 padding:8px;
 text-align:left;
 }

 th{
 background-color:#4CAF50;
 color:white;
 }

 h2{
```

```

 text-align:center;
 }
</style>

</head>

<body>

<h2>Technology Information</h2>

<table>

<tr>
 <th>ID</th>
 <th>Name</th>
 <th>Email</th>
 <th>Contact</th>
 <th>Technology</th>
 <th>Category</th>
 <th>Type</th>
 <th>Year</th>
 <th>Description</th>
</tr>

<?php
while($row = mysqli_fetch_assoc($result))
{
 ?>

<tr>
 <td><?php echo $row["TechnologyID"]; ?></td>
 <td><?php echo $row["Name"]; ?></td>
 <td><?php echo $row["Email"]; ?></td>
 <td><?php echo $row["Contact"]; ?></td>
 <td><?php echo $row["TechnologyName"]; ?></td>
 <td><?php echo $row["Category"]; ?></td>
 <td><?php echo $row["Type"]; ?></td>
 <td><?php echo $row["YearIntroduced"]; ?></td>
 <td><?php echo $row["Description"]; ?></td>
</tr>

<?php
}

mysqli_close($conn);

?>

</table>

</body>
</html>

```

### Question 3: Update Technology Information

## Create a PHP program to update the Category and Year Introduced of an existing technology record.

### Steps:

1. Create update.php.
2. Connect to TechnologyPortal database.
3. Accept TechnologyID and updated values.
4. Execute UPDATE query.
5. Display success message.

### PHP Code (update.php)

```
<?php

// Database Connection
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "TechnologyPortal";

// Create Connection
$conn = mysqli_connect($servername, $username, $password, $dbname);

// Check Connection
if (!$conn)
{
 die("Connection Failed: " . mysqli_connect_error());
}

// Check whether form is submitted
if(isset($_POST["update"]))
{
 $technologyID = $_POST["technologyID"];
 $category = $_POST["category"];
 $yearIntroduced = $_POST["yearIntroduced"];

 // Update Query
 $sql = "UPDATE Technology
 SET Category='$category',
 YearIntroduced='$yearIntroduced'
 WHERE TechnologyID='$technologyID'";

 if(mysqli_query($conn, $sql))
 {
 echo "<h2>Technology record updated successfully.</h2>";
 }
 else
 {
 echo "Error: " . mysqli_error($conn);
 }
}

mysqli_close($conn);

?>

<!DOCTYPE html>
<html>
```

```
<head>
 <title>Update Technology Record</title>

 <style>
 body{
 font-family: Arial, sans-serif;
 background:#f2f2f2;
 }

 .container{
 width:400px;
 margin:40px auto;
 background:#ffffff;
 padding:20px;
 border-radius:8px;
 box-shadow:0 0 10px gray;
 }

 h2{
 text-align:center;
 }

 input{
 width:100%;
 padding:8px;
 margin:8px 0;
 box-sizing:border-box;
 }

 input[type="submit"]{
 background:#4CAF50;
 color:white;
 border:none;
 cursor:pointer;
 }

 input[type="submit"]:hover{
 background:#45a049;
 }
 </style>
</head>

<body>

<div class="container">

<h2>Update Technology</h2>

<form method="post">

 <label>Technology ID</label>
 <input type="number" name="technologyID" required>

 <label>New Category</label>
 <input type="text" name="category" required>

 <label>Year Introduced</label>
 <input type="number" name="yearIntroduced" required>

 <input type="submit" name="update" value="Update Record">

</form>
```

```
</div>

</body>
</html>
```

#### Question 4: Delete Technology Information

Create a PHP program to delete a technology record from the **Technology** table using **TechnologyID**.

##### Steps:

1. Create delete.php.
2. Connect to TechnologyPortal database.
3. Accept TechnologyID.
4. Execute DELETE query.
5. Display success message.

#### Question 9: Create a PHP Program to Delete a Technology Record

Create a PHP program to delete a technology record from the **Technology** table using **TechnologyID**.

##### Steps

1. Create a file named **delete.php** inside the **TechnologyPortal** folder in **htdocs**.
2. Connect to the **TechnologyPortal** MySQL database.
3. Accept **TechnologyID** from the user.
4. Execute the **DELETE** query.
5. Display a success message after deleting the record.

##### PHP Code (delete.php)

```
<?php
```

```
// Database Connection
```

```
$servername = "localhost";
```

```
$username = "root";
```

```
$password = "";
```

```
$dbname = "TechnologyPortal";
```

```
// Create Connection
```

```
$conn = mysqli_connect($servername, $username, $password, $dbname);
```

```
// Check Connection
```

```
if (!$conn)
```

```
{
```

```
 die("Connection Failed: " . mysqli_connect_error());
```

```
}
```

```
// Check whether form is submitted
```

```
if(isset($_POST["delete"]))
```

```
{
```

```
 $technologyID = $_POST["technologyID"];
```

```
 // Delete Query
```

```
 $sql = "DELETE FROM Technology
```

```
 WHERE TechnologyID='$technologyID'";
```

```
 if(mysqli_query($conn, $sql))
```

```
 {
```

```
 echo "<h2>Technology record deleted successfully.</h2>";
```

```
 }
```

```
 else
```

```
 {
```

```
 echo "Error: " . mysqli_error($conn);
```

```
 }
```

```
}
```

```
mysqli_close($conn);
```

?>

<!DOCTYPE html>

<html>

<head>

<title>Delete Technology Record</title>

<style>

body{

font-family: Arial, sans-serif;

background:#f2f2f2;

}

.container{

width:400px;

margin:40px auto;

background:#ffffff;

padding:20px;

border-radius:8px;

box-shadow:0 0 10px gray;

}

h2{

text-align:center;

}

input{

width:100%;

padding:8px;

margin:8px 0;

```
 box-sizing: border-box;
}
```

```
input[type="submit"]{
 background: #dc3545;
 color: white;
 border: none;
 cursor: pointer;
}
```

```
input[type="submit"]:hover{
 background: #b02a37;
}
```

```
</style>
</head>
```

```
<body>
```

```
<div class="container">
```

```
<h2>Delete Technology Record</h2>
```

```
<form method="post">
```

```
<label>Technology ID</label>
```

```
<input type="number" name="technologyID" required>
```

```
<input type="submit" name="delete" value="Delete Record">
```

```
</form>
```

</div>

</body>

</html>

## 6. Result

Successfully performed all the questions.

## 7. Applications

- Data Management Systems
- ERP Applications
- Dynamic Websites

## 8. Practice Questions

1. Create a PHP program to search a technology by its name and display the corresponding record from the Technology table.
2. Create a PHP program to display only those technologies belonging to a selected category.

### Categories:

- Artificial Intelligence
- Cloud Computing
- Cyber Security
- IoT
- Blockchain

3. Create a PHP program to display the total number of technology records available in the Technology table.

## 9. Viva Questions

1. What is CRUD?
2. What is INSERT query?
3. What is UPDATE query?
4. What is DELETE query?
5. What is SELECT query?
6. What is MySQLi?
7. What is a database connection?
8. Why are CRUD operations important?
9. What is SQL injection?

10. How can database security be improved?